3CMa

Continuum Mechanics I

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Chapter 1

Tensors, Stress and Strain

1.1 Index notation

1.1.1 Vectors and Matrices

The expression u_i denotes the ith Cartesian component of the vector $\boldsymbol{u}$. Similarly, M_{ij} denotes the (i, j) component of the matrix M. The vector equation

$$\boldsymbol{c} = \boldsymbol{a} + \boldsymbol{b}$$

can be written as

$$c_i = a_i + b_i$$

or

$$c_j = a_j + b_j.$$

The choice of index labelling the component of the vector is arbitrary. Similarly,

$$P = M + N$$

can be written as

$$P_{ij} = M_{ij} + N_{ij}.$$

1.1.2 Scalar Product

The scalar product of two vectors is given by

$$\boldsymbol{u} \cdot \boldsymbol{v} = \sum_{i=1}^{3} u_i v_i.$$

The index notation omits the summation sign $\sum$ for repeated indices

$$\boxed{\boldsymbol{u} \cdot \boldsymbol{v} = u_i v_i.}$$

Example 1.1.1. *Rewrite $a_i b_j c_i d_j$ in terms of scalar products.*

$$\begin{aligned} a_i b_j c_i d_j &= \sum_{i=1}^{3}\sum_{j=1}^{3} a_i b_j c_i d_j \\ &= \left(\sum_{i=1}^{3} a_i c_i\right)\left(\sum_{j=1}^{3} b_j d_j\right) \\ &= (\boldsymbol{a}\cdot\boldsymbol{c})\,(\boldsymbol{b}\cdot\boldsymbol{d})\,. \end{aligned}$$

Example 1.1.2. *Is $a_q b_p c_q d_p$ the same as $a_i b_j c_i d_j$?*

$$\begin{aligned} a_q b_p c_q d_p &= \sum_{p=1}^{3}\sum_{q=1}^{3} a_q b_p c_q d_p \\ &= \left(\sum_{p=1}^{3} b_p d_p\right)\left(\sum_{q=1}^{3} a_q c_q\right) \\ &= (\boldsymbol{b}\cdot\boldsymbol{d})\,(\boldsymbol{a}\cdot\boldsymbol{c}) \\ &= (\boldsymbol{a}\cdot\boldsymbol{c})\,(\boldsymbol{b}\cdot\boldsymbol{d})\,. \end{aligned}$$

Repeated indices are dummy variables.

Example 1.1.3. *Rewrite $a_q b_q c_p d_p$ in terms of scalar products.*

$$\begin{aligned} a_q b_q c_p d_p &= \sum_{p=1}^{3}\sum_{q=1}^{3} a_q b_q c_p d_p \\ &= \left(\sum_{p=1}^{3} c_p d_p\right)\left(\sum_{q=1}^{3} a_q b_q\right) \\ &= (\boldsymbol{c}\cdot\boldsymbol{d})\,(\boldsymbol{a}\cdot\boldsymbol{b})\,. \end{aligned}$$

We note that this result is not the same as in the previous two examples.

Rule 1 Any index cannot appear more than twice within a single expression.

Example 1.1.4. *The expression $a_p b_p c_p d_p$ is unacceptable.*

Rule 2 Ensure all unrepeated indices balance.

Rule 2 may be broken, but it is extremely unlikely.

Example 1.1.5. *Write the equation $(\boldsymbol{a}\cdot\boldsymbol{b})\boldsymbol{c} = \boldsymbol{b}$ in index notation.*
We have $\boldsymbol{a}\cdot\boldsymbol{b} = a_i b_i$ so that the pth component $p \in \{1,2,3\}$ is

$$a_i b_i c_p = b_p.$$

Example 1.1.6. *Determine the meaning of the expression*

$$a_i b_i c_q = b_p.$$

This expression states that the qth component of $(\boldsymbol{a}\cdot\boldsymbol{b})\boldsymbol{c}$ *is equal to the pth component of* $\boldsymbol{b}$*. This is* 9 *equations for* $p,\ q \in \{1,2,3\}$*. This example breaks rule 2.*

1.1.3 Matrix Multiplication

The matrix equation $P = MN$ may be written in index notation as

$$P_{ij} = M_{ik}N_{kj}.$$

P_{ij} is the ith component of P obtained by summing the ith row of M and jth column of N.

$$\begin{pmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{pmatrix} \begin{pmatrix} N_{11} & N_{12} & N_{13} \\ N_{21} & N_{22} & N_{23} \\ N_{31} & N_{32} & N_{33} \end{pmatrix} = P.$$

In other words,

$$P_{11} = M_{11}N_{11} + M_{12}N_{21} + M_{13}N_{31}$$

and so on.

Additional examples

- Rewrite $(1 + (\boldsymbol{a}\cdot\boldsymbol{b})^2)\boldsymbol{u} = (\boldsymbol{c}\cdot\boldsymbol{c})\boldsymbol{b}$ in index notation.
- Does $M_{ij} = a_i b_j$ make sense?
- Does $a_i b_j c_k$ make sense?
- Does $a_i b_i c_i$ make sense?

1.1.4 Div and Grad

Definition 1.1.7. *We define the following notation*

$$\partial_1 = \frac{\partial}{\partial x_1} = \frac{\partial}{\partial x}, \qquad \partial_2 = \frac{\partial}{\partial x_2} = \frac{\partial}{\partial y}, \qquad \partial_3 = \frac{\partial}{\partial x_3} = \frac{\partial}{\partial z}.$$

Definition 1.1.8. *Let* ϕ *be a scalar field. The* i*th component of* $\boldsymbol{\nabla}\phi$ *is given by*

$$[\boldsymbol{\nabla}\phi]_i = \frac{\partial \phi}{\partial x_i} = \partial_i \phi.$$

Definition 1.1.9. *Let* $\boldsymbol{u}$ *be a vector field. The divergence is given by*

$$\boldsymbol{\nabla}\cdot\boldsymbol{u} = \frac{\partial u_i}{\partial x_i} = \partial_i u_i.$$

Example 1.1.10. *Use index notation to show that* $\boldsymbol{\nabla}(\phi\psi) = \phi\boldsymbol{\nabla}\psi + \psi\boldsymbol{\nabla}\phi$*, where* ϕ *and* ψ *are scalar fields.*
Consider the ith component

$$\begin{aligned}[\boldsymbol{\nabla}(\phi\psi)]_i &= \frac{\partial}{\partial x_i}(\phi\psi) \\ &= \phi\frac{\partial\psi}{\partial x_i} + \psi\frac{\partial\phi}{\partial x_i} \\ &= \phi[\boldsymbol{\nabla}\psi]_i + \psi[\boldsymbol{\nabla}\phi]_i \\ &= [\phi\boldsymbol{\nabla}\psi + \psi\boldsymbol{\nabla}\phi]_i\end{aligned}$$

Thus, $\boldsymbol{\nabla}(\phi\psi) = \phi\boldsymbol{\nabla}\psi + \psi\boldsymbol{\nabla}\phi$.

Example 1.1.11. *Use index notation to show that* $\boldsymbol{\nabla}\cdot(\phi\boldsymbol{u}) = \phi\boldsymbol{\nabla}\cdot\boldsymbol{u} + \boldsymbol{u}\cdot\boldsymbol{\nabla}\phi$*, where* ϕ *is a scalar field and* $\boldsymbol{u}$ *is a vector field.*
Apply the product rule

$$\begin{aligned}\boldsymbol{\nabla}\cdot(\phi\boldsymbol{u}) &= \frac{\partial}{\partial x_i}(\phi u_i) \\ &= \phi\frac{\partial u_i}{\partial x_i} + u_i\frac{\partial\phi}{\partial x_i} \\ &= \phi\boldsymbol{\nabla}\cdot\boldsymbol{u} + \boldsymbol{u}\cdot\boldsymbol{\nabla}\phi\end{aligned}$$

and hence the result.

Definition 1.1.12. *The Kronecker delta is defined to be*

$$\delta_{ij} = \begin{cases}1 & \text{if } i = j, \\ 0 & \text{if } i \neq j.\end{cases}$$

Notes

1. $\delta_{ij} = \delta_{ji}$; that is, δ_{ij} is symmetric.
2. δ_{ij} is the analogue of the identity matrix in index notation.

Definition 1.1.13. P_{ij} *is antisymmetric if* $P_{ij} = -P_{ji}$.

Example 1.1.14. *Simplify the expression* $\delta_{pq}a_q$.

$$\delta_{pq}a_q = \delta_{p1}a_1 + \delta_{p2}a_2 + \delta_{p3}a_3 = a_p.$$

Example 1.1.15. *Simplify the expression* $\delta_{rs}u_sv_r$.

$$\begin{aligned}\delta_{rs}u_sv_r &= (\delta_{rs}u_s)\,v_r \\ &= (\delta_{r1}u_1 + \delta_{r2}u_2 + \delta_{r3}u_3)\,v_r = u_rv_r = \mathbf{u}\cdot\mathbf{v}\end{aligned}$$

Example 1.1.16. *Simplify the expression* $\delta_{\xi\xi}$.

$$\delta_{\xi\xi} = \delta_{11} + \delta_{22} + \delta_{33} = 3$$

Example 1.1.17. *Simplify the expression* $\delta_{ij}\delta_{pq}a_p b_i c_q$.

$$\delta_{ij}\delta_{pq}a_p b_i c_q = \delta_{ij}b_i\left(\delta_{pq}a_p c_q\right) = \delta_{ij}b_i(\mathbf{a}\cdot\mathbf{c}) = (\mathbf{a}\cdot\mathbf{c})b_j$$

Example 1.1.18. *Simplify the expression* $\delta_{ij}\left(a_i b_j - a_j b_i\right)$.

There are many ways of showing that this is zero. The most important, in that this observation can be applied in many situations, is the following. δ_{ij} *is symmetric in* (i,j)*, while* $(a_i b_j - b_i a_j)$ *is antisymmetric in* (i,j)*. In more detail ... let* $P_{ij} \overset{def}{=} (a_i b_j - a_j b_i)$*. Then*

$$\begin{aligned} \delta_{ij}P_{ij} &= \delta_{ji}P_{ij} && as\ \ \delta_{ij} = \delta_{ji} \\ &= -\delta_{ji}P_{ji} && as\ P_{ij} = -P_{ji} \\ &= -\delta_{ij}P_{ij} \end{aligned}$$

where the last line follows since as i, j *are dummy indices. They are summed and so can be replaced with other symbols, without changing the meaning of the index notation expression under consideration. Here we have replaced* (i,j) *with* (j,i)*.*

Notes

1. u_i, a_j, b_k... are numbers; we can write $a_i b_j = b_j a_i$ just as we can write $3 \times 4 = 4 \times 3$, and we can swap the order of u_i, a_j, b_k... at will within an index notation expression.

2. symmetry-anti-symmetry arguments can be used to deduce, by inspection, that various expressions are zero. In particular, suppose we have $P_{...ij...}R_{...ij...}$ where $P_{...ij...} = P_{...ji...}$, and $R_{...ij...} = -R_{...ji...}$. Then $P_{...ij...}R_{...ij...} = 0$ by application of the reasoning in (v) above.

1.1.5 The Permutation Symbol

Definition 1.1.19. *The permutation symbol is defined to be*

$$\varepsilon_{ijk} = \begin{cases} 1 & \text{if } (i,j,k) = (1,2,3) \text{ or } (3,1,2) \text{ or } (2,3,1), \\ -1 & \text{if } (i,j,k) = (1,3,2) \text{ or } (2,1,3) \text{ or } (3,2,1), \\ 0 & \text{if any of } i,j \text{ or } k \text{ are equal.} \end{cases}$$

Notes

1. The permutation symbol represents notation that enables us to easily manipulate cross products and curls.

2. The permutation symbol is neglected if two indices are swapped. For example,

$$\varepsilon_{ijk} = -\varepsilon_{jik} = -\varepsilon_{ikj} = -\varepsilon_{kji}.$$

3. The permutation symbol is unchanged if the indices are shifted to the right, putting the third index first. For example,

$$\varepsilon_{ijk} = \varepsilon_{kij}.$$

Example 1.1.20. *Show that* $[\boldsymbol{u} \wedge \boldsymbol{v}]_i = \varepsilon_{ijk} u_j v_k$.

We consider the first component

$$\begin{aligned} \varepsilon_{1jk} u_j v_k &= \varepsilon_{111} u_1 v_1 + \varepsilon_{112} u_1 v_2 + \varepsilon_{113} u_1 v_3 \\ &\quad + \varepsilon_{121} u_2 v_1 + \varepsilon_{122} u_2 v_2 + \varepsilon_{123} u_2 v_3 \\ &\quad + \varepsilon_{131} u_3 v_1 + \varepsilon_{132} u_3 v_2 + \varepsilon_{133} u_3 v_3 \\ &= u_2 v_3 - u_3 v_2 = [\boldsymbol{u} \wedge \boldsymbol{v}]_1. \end{aligned}$$

Similarly, we calculate the second and third components.

Exercise

Show that $[\boldsymbol{\nabla} \wedge \boldsymbol{v}]_i = \varepsilon_{ijk} \dfrac{\partial v_k}{\partial x_j} = \varepsilon_{ijk} \partial_j v_k$.

Example 1.1.21. *Use index notation to show* $\mathbf{a} \wedge \mathbf{b} = -\mathbf{b} \wedge \mathbf{a}$.

$$\begin{aligned} [\mathbf{a} \wedge \mathbf{b}]_i &= \varepsilon_{ijk} a_j b_k && \textit{by definition} \\ &= \varepsilon_{ijk} b_k a_j && \textit{as we can swap } a_j, b_k \\ &= -\varepsilon_{ikj} b_k a_j && \textit{by antisymmetry of indices for } \varepsilon_{ijk} \\ &= -[\mathbf{b} \wedge \mathbf{a}]_i && \textit{by definition} \end{aligned}$$

Example 1.1.22. *Show that* $(\mathbf{a} \cdot (\mathbf{b} \wedge \mathbf{c})) = \varepsilon_{ijk} a_i b_j c_k$. *Hence show that* $(\mathbf{a} \cdot (\mathbf{b} \wedge \mathbf{c})) = -(\mathbf{c} \cdot (\mathbf{b} \wedge \mathbf{a}))$.

$$\begin{aligned} \mathbf{a} \cdot (\mathbf{b} \wedge \mathbf{c}) &= a_i (\mathbf{b} \wedge \mathbf{c})_i = a_i \varepsilon_{ijk} b_j c_k = \varepsilon_{ijk} a_i b_j c_k \\ &= c_k (\varepsilon_{ijk} b_j a_i) = c_k (\varepsilon_{kij} b_j a_i) \\ &= c_k (-\varepsilon_{kji} b_j a_i) = c_k (-\varepsilon_{kpq} b_p a_q) = -\mathbf{c} \cdot (\mathbf{b} \wedge \mathbf{a}) \end{aligned}$$

Example 1.1.23. *Simplify* $\varepsilon_{ijk} \delta_{jk}$. *Simplify* $\varepsilon_{ijk} a_j a_k$.

You should be able to see that both expressions are zero, by using the antisymmetry properties of ε_{ijk}. *It is antisymmetric in all its indices, and so* $\varepsilon_{ijk} = -\varepsilon_{ikj}$, *while* $a_j a_k = a_k a_j$, $\delta_{jk} = \delta_{kj}$. *Hence both expressions are zero, by the symmetry, antisymmetry arguments.*

Example 1.1.24. *Simplify* $\nabla \cdot (\nabla \wedge \mathbf{u})$.

$$\nabla \cdot (\nabla \wedge \mathbf{u}) = \partial_i(\varepsilon_{ijk}\partial_j u_k) = \varepsilon_{ijk}\partial_i\partial_j u_k = 0$$

and the final expression is zero by symmetry, antisymmetry arguments.

Example 1.1.25. *Simplify* $\nabla \wedge (\nabla\varphi)$, *where* φ *is a scalar field.*

$$[\nabla \wedge (\nabla\varphi)]_i = \varepsilon_{ijk}\partial_j(\partial_k\varphi) = \varepsilon_{ijk}\partial_j\partial_k\varphi = 0$$

and the final expression is zero by symmetry, antisymmetry arguments.

Example 1.1.26. *Show that* $[\nabla \wedge (\nabla \wedge \mathbf{u})]_i = \varepsilon_{kij}\varepsilon_{kpq}\partial_j\partial_p u_q$.

$$[\nabla \wedge \mathbf{u}]_k = \varepsilon_{kpq}\partial_p u_q$$

Thus

$$[\nabla \wedge (\nabla \wedge \mathbf{u})]_i = \varepsilon_{ijk}\partial_j(\varepsilon_{kpq}\partial_p u_q) = \varepsilon_{ijk}\varepsilon_{kpq}\partial_j\partial_p u_q = \varepsilon_{kij}\varepsilon_{kpq}\partial_j\partial_p u_q$$

Notes

1. We see, in a simple setting, a number of techniques typically used in index notation ... the antisymmetry of the indices of ε_{ijk}; the fact we can swap $a_j,\ b_k$ in an index notation expression; the fact we can use any symbol we choose for dummy indices; and the symmetry, antisymmetry arguments used in examples 3, 4 and 5.

2. We expressed $(\nabla \wedge (\nabla \wedge \mathbf{u}))_i$ in terms of the permutation symbol. Thus to use the index notation for simplifying multiple applications of curl on a vector field, we need an expression for $\varepsilon_{kij}\varepsilon_{kpq}$.

Exercise

Show that $\varepsilon_{kij}\varepsilon_{kpq} = \delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp}$. We note that the 4 free indices correspond to $3^4 = 81$ equations.

Example 1.1.27. *Show that* $\boldsymbol{\nabla} \wedge (\boldsymbol{\nabla} \wedge \mathbf{u}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \boldsymbol{u}) - \nabla^2\boldsymbol{u}$.

$$\begin{aligned}
[\boldsymbol{\nabla} \wedge (\boldsymbol{\nabla} \wedge \mathbf{u})]_i &= \varepsilon_{kij}\varepsilon_{kpq}\frac{\partial^2 u_q}{\partial x_j \partial x_p} \\
&= (\delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp})\frac{\partial^2 u_q}{\partial x_j \partial x_p} \\
&= \frac{\partial^2 u_j}{\partial x_j \partial x_i} - \frac{\partial^2 u_i}{\partial x_j \partial x_j} \\
&= \frac{\partial}{\partial x_i}(\boldsymbol{\nabla} \cdot \boldsymbol{u}) - [\nabla^2\boldsymbol{u}]_i \\
&= [\boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \boldsymbol{u}) - \nabla^2\boldsymbol{u}]_i
\end{aligned}$$

Example 1.1.28. *Show that* $\boldsymbol{\nabla} \wedge (\boldsymbol{u} \wedge \boldsymbol{v}) = \boldsymbol{u}(\boldsymbol{\nabla} \cdot \boldsymbol{v}) + \boldsymbol{v} \cdot \boldsymbol{\nabla}\boldsymbol{u} - \boldsymbol{u} \cdot \boldsymbol{\nabla}\boldsymbol{v} - \boldsymbol{v}(\boldsymbol{\nabla} \cdot \boldsymbol{u})$.

$$\begin{aligned}
[\boldsymbol{\nabla} \wedge (\boldsymbol{u} \wedge \mathbf{v})]_i &= \varepsilon_{ijk} \frac{\partial}{\partial x_j}(\varepsilon_{kpq} u_p v_q) \\
&= \varepsilon_{kij}\varepsilon_{kpq} \frac{\partial}{\partial x_j}(u_p v_q) \\
&= (\delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp}) \frac{\partial}{\partial x_j}(u_p v_q) \\
&= \frac{\partial}{\partial x_j}(u_i v_j) - \frac{\partial}{\partial x_j}(u_j v_i) \\
&= u_i \frac{\partial v_j}{\partial x_j} + v_j \frac{\partial u_i}{\partial x_j} - u_j \frac{\partial v_i}{\partial x_j} - v_i \frac{\partial u_j}{\partial x_j} \\
&= [\boldsymbol{u}(\boldsymbol{\nabla} \cdot \boldsymbol{v}) + \boldsymbol{v} \cdot \boldsymbol{\nabla}\boldsymbol{u} - \boldsymbol{u} \cdot \boldsymbol{\nabla}\boldsymbol{v} - \boldsymbol{v}(\boldsymbol{\nabla} \cdot \boldsymbol{u})]_i
\end{aligned}$$

1.2 An Introduction to Tensors

Equations of mechanics and physics are often expressed in vector notation, for example *Newton's Second Law* is commonly expressed as

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(m\frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t}\right) = \boldsymbol{F}(t).$$

The underlying reason for writing equations in vector form is the assumption that the laws of nature do not depend on the choice of coordinate system. However physical quantities cannot all be classified as vector or scalars, and the notion of *Tensors* is required.

The tensor enables us to extend the notion that equations are independent of the choice of coordinate system. *Cartesian Tensors* are associated with cartesian frames of reference in Euclidean space. We shall not deal with general tensor theory and the associated non-Cartesian frames of reference. Tensors will be dependent variables in our discussion of the theory of fluid mechanics and elasticity.

1.2.1 Cartesian Tensors

Consider a rotation of the set of axes, $\boldsymbol{i}_1, \boldsymbol{i}_2, \boldsymbol{i}_3$ through to $\boldsymbol{i}'_1, \boldsymbol{i}'_2, \boldsymbol{i}'_3$, then the same vector can be written in the form $\boldsymbol{x} = x_k \boldsymbol{i}_k$ and $\boldsymbol{x}' = x'_m \boldsymbol{i}'_m$. Now by the Covariance principle these represent the same point and thus

$$x_k \boldsymbol{i}_k = x'_m \boldsymbol{i}'_m. \tag{1.1}$$

Taking Scalar products with $\boldsymbol{i}'_p$ yields

$$x_k \boldsymbol{i}_k . \boldsymbol{i}'_p = x'_m \boldsymbol{i}'_m . \boldsymbol{i}'_p = x'_m \delta_{mp} = x'_p,$$

which by defining $l_{pk} = \boldsymbol{i}'_p . \boldsymbol{i}_k$, the direction cosine between $\boldsymbol{i}'_p$ and $\boldsymbol{i}_k$ yields

$$x'_p = l_{pk} x_k. \tag{1.2}$$

If however we take scalar products of equation (1.1), with $\boldsymbol{i}_p$ then we obtain

$$\begin{aligned} x_k \boldsymbol{i}_k.\boldsymbol{i}_p &= x'_m \boldsymbol{i}'_m.\boldsymbol{i}_p, \\ x_k \delta_{kp} &= x'_m \boldsymbol{i}'_m.\boldsymbol{i}_p, \\ \Rightarrow \quad x_p &= x'_m l_{mp}. \end{aligned} \tag{1.3}$$

From equation (1.2) we have

$$\boldsymbol{x}' = \begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \end{pmatrix} = \begin{pmatrix} l_{11} & l_{12} & l_{13} \\ l_{21} & l_{22} & l_{23} \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = L\boldsymbol{x}.$$

or in matrix notation $\boldsymbol{x}' = L\boldsymbol{x}$ and similarly from equation 1.3 we have $\boldsymbol{x} = L^T\boldsymbol{x}'$, so we deduce that $\boldsymbol{x}' = L(L^T\boldsymbol{x}')$ so that $LL^T = L^TL = I$ and therefore L represents an orthogonal transformation. Another way to see this is that $l_{pk}l_{mk} = \delta_{pm}$ and that $l_{kp}l_{km} = \delta_{pm}$.

Definition 1.2.1 (First Order Cartesian Tensors)**.** *Any quantity x_k which satisfies*

$$x'_p = l_{pk}x_k.$$

Definition 1.2.2 (Second Order Cartesian Tensors)**.** *Any quantity a_{ij} which transforms as*

$$a'_{mn} = l_{mi}l_{nj}a_{ij}.$$

1.2.2 Some Properties of Tensors

1. Multiplying a tensor by a scalar, obtains a tensor of the same order.

2. Two tensors of the same order are added (subtracted) by adding (subtracting) the corresponding components

3. Two tensors can be multiplied together, $a_{ijk}b_{lm} = c_{ijklm}$, this is call the outer product.

4. If two indices are equal, in the outer product we can obtain tensors two orders lower $a_{isk}b_{jk} = d_{isj}$. A specific example is the scalar product, which given two vectors returns a scalar.

5. If $a_{ij} = a_{ji}$, then a_{ij} is a symmetric tensor with 6 independent components.

6. If $a_{ij} = -a_{ji}$, then a_{ij} is an anti-symmetric Tensor with 3 independent components.

Example 1.2.3. *Show that δ_{ij} is a tensor, that is $\delta'_{mn} = l_{mi}l_{nj}\delta_{ij}$.*
If $i \neq j$, then $l_{mi}l_{nj}\delta_{ij} = 0$, if $i = j$ then $l_{mi}l_{nj}\delta_{ij} = l_{mj}l_{nj} = \delta_{mn}$ by orthogonality and hence $\delta'_{mn} = \delta_{mn}$, the Kronecker delta is actually transformed to itself.

1.2.3 The Continuum Hypothesis

We assume that the highly discontinuous structures of real materials may be replaced by a smoothed out hypothetical continuum, every potion of this material (no matter how small) having the same macroscopic physical properties as the bulk

Example 1.2.4. *Experiments indicate that the pressure p, density ρ and temperature T of many gasses and liquids are related through a constitutive equation of the form $p = f(\rho, T)$. This equation is called the equation of state. This equation remains valid over a wide range of sample states. Continuum mechanics assumes the relationship remains valid even for infinitesimal volumes.*

1.2.4 Eulerian and Lagrangian Coordinates

Definition 1.2.5. *A* Material Point *is a mathematical entity, within a continuum and is not to be confused with atoms or molecules.*

Suppose initially at $t = 0$ the *Material points* of a continuous media occupy a volume $v(0)$ and after some time t a volume $v(t)$.

The Lagrangian description of motion is

$$x_1 = x_1(X_1, X_2, X_3, t),$$
$$x_2 = x_2(X_1, X_2, X_3, t),$$
$$x_3 = x_3(X_1, X_2, X_3, t),$$

Velocity in the Lagrangian coordinate system is given by

$$\boldsymbol{V}(X_1, X_2, X_3, t) = \frac{\partial}{\partial t}\boldsymbol{x}(X_1, X_2, X_3, t).$$

Velocity in the Eulerian Coordinate system is given by

$$\boldsymbol{v}(x_1, x_2, x_3, t) = \boldsymbol{V}\left(X_1(x_1, x_2, x_3, t), X_2(x_1, x_2, x_3, t), X_3(x_1, x_2, x_3, t), t\right).$$

We shall adopt the convention of capital letters for the Lagrangian coordinates and lower cases letters for Eulerian Coordinates.

$$\text{Lagrangian} \quad \boldsymbol{X} = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}; \qquad \text{Eulerian} \quad \boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

The mapping $\boldsymbol{x} = \boldsymbol{x}(\boldsymbol{X}, t)$ is bijective and the inverse is given by

$$\boldsymbol{X} = \boldsymbol{X}(\boldsymbol{x}, t). \tag{1.4}$$

For a differentiable function $\boldsymbol{x}(\boldsymbol{X}, t)$, a necessary condition for the existence of the unique solution of equation (1.4) is the non-vanishing of the Jacobian, explicitly

$$J = \frac{\partial(x_1, x_2, x_3)}{\partial(X_1, X_2, X_3)} = \begin{vmatrix} \dfrac{\partial x_1}{\partial X_1} & \dfrac{\partial x_1}{\partial X_2} & \dfrac{\partial x_1}{\partial X_3} \\ \dfrac{\partial x_2}{\partial X_1} & \dfrac{\partial x_2}{\partial X_2} & \dfrac{\partial x_2}{\partial X_3} \\ \dfrac{\partial x_3}{\partial X_1} & \dfrac{\partial x_3}{\partial X_2} & \dfrac{\partial x_3}{\partial X_3} \end{vmatrix} \neq 0.$$

and that the partial derivatives $\partial x_i / \partial X_j$ are continuous. This follows from the 'Implicit Function Theorem'. These assumption reflect the indestructible nature of matter, that is to say that there is no region of finite volume can be deformed into a zero or infinite volume.

1.3 Strain

In rigid body dynamics, there is no relative motion between material points, in general however this is not true, there is compression and distortion of material. These effects may vary throughout the continuum. A description of deformation is necessary at each point of the continuum, the question is 'how to describe such deformations?'.

1.3.1 Analysis of Strain

The nature of the deformation of the material in the neighborhood of a particle labelled $\boldsymbol{X}$ is made clear by examining the behaviour of all line vectors connecting $\boldsymbol{X}$ with $\boldsymbol{X} + \mathrm{d}\boldsymbol{X}$. At time t this line has moved between the point $\boldsymbol{x}(\boldsymbol{X}, t)$ and $\boldsymbol{x}(\boldsymbol{X} + \mathrm{d}\boldsymbol{X}, t)$, such that

$$\mathrm{d}\boldsymbol{x} = \boldsymbol{x}(\boldsymbol{X} + \mathrm{d}\boldsymbol{X}, t) - \boldsymbol{x}(\boldsymbol{X}, t)$$

It is the nature of this transformation which determines the local deformation and not the translation of the vector line segment. We denote the square of the line vector $\mathrm{d}\boldsymbol{X}$ as $\mathrm{d}S^2 = \mathrm{d}X_K\,\mathrm{d}X_K$ and the square of the length of the line vector $\mathrm{d}\boldsymbol{x}$ as $\mathrm{d}s^2 = \mathrm{d}x_k\,\mathrm{d}x_k$.

We consider the difference in length

$$\mathrm{d}s^2 - \mathrm{d}S^2 = \mathrm{d}x_k\,\mathrm{d}x_k - \mathrm{d}X_K\,\mathrm{d}X_K.$$

This quantity is determined solely by the deformation and neither by the rotation nor other rigid body motion.

Example 1.3.1. *Consider a rigid body motion with velocity v in the X_1 direction, then*

$$\begin{aligned} x_1 &= X_1 + vt, & x_2 &= X_2, & x_3 &= X_3, \\ \mathrm{d}x_1 &= \mathrm{d}X_1, & \mathrm{d}x_2 &= \mathrm{d}X_2, & \mathrm{d}x_3 &= \mathrm{d}X_3. \end{aligned}$$

so $\mathrm{d}s^2 - \mathrm{d}S^2 = \mathrm{d}x_k\,\mathrm{d}x_k - \mathrm{d}X_K\,\mathrm{d}X_K = 0.$

Example 1.3.2. *Consider a 90° clockwise rotation about the x_3 axis, then*

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}.$$

so

$$\begin{aligned} \mathrm{d}x_1 &= \mathrm{d}X_2, \\ \mathrm{d}x_2 &= -\mathrm{d}X_1, \\ \mathrm{d}x_3 &= \mathrm{d}X_3. \end{aligned}$$

so $\mathrm{d}s^2 - \mathrm{d}S^2 = \mathrm{d}x_k\,\mathrm{d}x_k - \mathrm{d}X_K\,\mathrm{d}X_K = 0.$

Example 1.3.3. *Consider a body stretched in the X_1 direction ($\lambda \neq 1$)*

$$x_1 = \lambda X_1, \qquad x_2 = X_3, \qquad x_3 = X_3;$$

$$\mathrm{d}x_1 = \lambda\, \mathrm{d}X_1, \qquad \mathrm{d}x_2 = \mathrm{d}X_2, \qquad \mathrm{d}x_3 = \mathrm{d}X_3$$

$$\begin{aligned} \mathrm{d}s^2 - \mathrm{d}S^2 &= \mathrm{d}x_k\, \mathrm{d}x_k - \mathrm{d}X_K\, \mathrm{d}X_K, \\ &= \mathrm{d}x_1\ \mathrm{d}x_1 - \mathrm{d}X_1\, \mathrm{d}X_1, && \textit{All other terms cancel} \\ &= \lambda^2\, \mathrm{d}X_1 - \mathrm{d}X_1^2, \\ &= (\lambda^2 - 1)\, \mathrm{d}X_1^2 \neq 0. \end{aligned}$$

The quantity $\mathrm{d}s^2 - \mathrm{d}S^2$ will allow us to define the strain tensor in terms of deformation in this case $E_{11} = \frac{1}{2}(\lambda^2 - 1)$ is the only non-zero component of the stain tensor.

Recall that if $x_k = x_k(\boldsymbol{X}, t)$, then

$$\begin{aligned} \mathrm{d}x_k &= x_k(\boldsymbol{X} + \mathrm{d}\boldsymbol{X}, t) - x_k(\boldsymbol{X}, t), \\ &= \frac{\partial x_k}{\partial X_K}\, \mathrm{d}X_K = x_{k,K}\, \mathrm{d}X_K. \end{aligned} \tag{1.5}$$

Also that if $X_k = X_k(\boldsymbol{x}, t)$,then

$$\begin{aligned} \mathrm{d}X_K &= X_K(\boldsymbol{x} + \mathrm{d}\boldsymbol{x}, t) - X_K(\boldsymbol{x}, t), \\ &= \frac{\partial X_K}{\partial x_k}\, \mathrm{d}x_k = X_{K,k}\ \mathrm{d}x_k. \end{aligned} \tag{1.6}$$

Note now $\mathrm{d}x_k = x_{k,K}\, \mathrm{d}X_K = x_{k,K} X_{K,l}\ \mathrm{d}x_l \ \Rightarrow x_{k,K} X_{K,l} = \delta_{kl}$. Similarly $X_{K,k} x_{k,M} = \delta_{KM}$.

We now use these in our expression for $\mathrm{d}s^2 - \mathrm{d}S^2$.

$$\begin{aligned} \mathrm{d}s^2 - \mathrm{d}S^2 &= \mathrm{d}x_k\, \mathrm{d}x_k - \mathrm{d}X_K\, \mathrm{d}X_K, \\ &= x_{k,K} x_{k,L}\, \mathrm{d}X_K\, \mathrm{d}X_L - \mathrm{d}X_K\, \mathrm{d}X_K, \\ &= x_{k,K} x_{k,L}\, \mathrm{d}X_K\, \mathrm{d}X_L - \delta_{KL}\, \mathrm{d}X_K\, \mathrm{d}X_L, \\ &= (x_{k,K} x_{k,L} - \delta_{KL})\, \mathrm{d}X_K\, \mathrm{d}X_L, \\ &= (C_{KL} - \delta_{KL})\, \mathrm{d}X_K\, \mathrm{d}X_L. \end{aligned}$$

Here $C_{KL} = x_{k,K} x_{k,L}$ is Green's deformation tensor and we have the following definition.

Definition 1.3.4 (Lagrangian Strain Tensor)**.**

$$\boxed{2E_{KL} = x_{k,K} x_{k,L} - \delta_{KL}}$$

Alternatively we could write

$$\begin{aligned} \mathrm{d}s^2 - \mathrm{d}S^2 &= \mathrm{d}x_k\,\mathrm{d}x_k - \mathrm{d}X_K\,\mathrm{d}X_K, \\ &= \mathrm{d}x_k\,\mathrm{d}x_k - X_{K,k}X_{K,l}\;\mathrm{d}x_k\;\mathrm{d}x_l, \\ &= \delta_{kl}\;\mathrm{d}x_k\;\mathrm{d}x_l - X_{K,k}X_{K,l}\;\mathrm{d}x_k\;\mathrm{d}x_l, \\ &= \;\mathrm{d}x_k\;\mathrm{d}x_l(\delta_{kl} - X_{K,k}X_{K,l}), \\ &= \;\mathrm{d}x_k\;\mathrm{d}x_l(\delta_{kl} - c_{kl}). \end{aligned}$$

Here $c_{kl} = X_{K,k}X_{K,l}$ is Cauchy's deformation tensor, and we have the following definition.

Definition 1.3.5 (Eulerian Strain Tensor)**.**

$$\boxed{2e_{kl} = \delta_{kl} - X_{K,k}X_{K,l}}$$

The Lagrangian Strain Tensor plays a central role in elasticity. The deformation rate tensor plays an equivalent role in fluid mechanics. We will derive the deformation rate tensor in terms of the Eulerian Strain Tensor.

Example 1.3.6. *Consider a body stretched in the X_1 direction, as in example 1.3.3*

$$x_1 = \lambda X_1; \qquad x_2 = X_2; \qquad x_3 = X_3.$$

Then the components of the Lagrangian Strain Tensor are

$$\begin{aligned} 2E_{11} = \frac{\partial x_k}{\partial X_1}.\frac{\partial x_k}{\partial X_1} - \delta_{11}, &= \left(\frac{\partial x_1}{\partial X_1}\right)^2 + \left(\frac{\partial x_2}{\partial X_1}\right)^2 + \left(\frac{\partial x_3}{\partial X_1}\right)^2 - \delta_{11}, \\ &= \lambda^2 + 0 + 0 - 1, \\ &= \lambda^2 - 1. \end{aligned}$$

$$\begin{aligned} 2E_{22} = \frac{\partial x_k}{\partial X_2}.\frac{\partial x_k}{\partial X_2} - \delta_{22}, &= \left(\frac{\partial x_1}{\partial X_2}\right)^2 + \left(\frac{\partial x_2}{\partial X_2}\right)^2 + \left(\frac{\partial x_3}{\partial X_2}\right)^2 - \delta_{22}. \\ &= 0 + 1 + 0 - 1 = 0. \end{aligned}$$

$$\begin{aligned} 2E_{33} = \frac{\partial x_k}{\partial X_3}.\frac{\partial x_k}{\partial X_3} - \delta_{33}, &= \left(\frac{\partial x_1}{\partial X_3}\right)^2 + \left(\frac{\partial x_2}{\partial X_3}\right)^2 + \left(\frac{\partial x_3}{\partial X_3}\right)^2 - \delta_{33}, \\ &= 0 + 0 + 0 + 1 - 1 = 0. \end{aligned}$$

$$\begin{aligned} 2E_{12} = \frac{\partial x_k}{\partial X_1}.\frac{\partial x_k}{\partial X_2} - \delta_{12}, &= \frac{\partial x_1}{\partial X_1}\frac{\partial x_1}{\partial X_2} + \frac{\partial x_2}{\partial X_1}\frac{\partial x_2}{\partial X_2} + \frac{\partial x_3}{\partial X_1}\frac{\partial x_3}{\partial X_2}, \\ &= 0. \end{aligned}$$

Similarly $E_{13} = E_{12} = E_{23} = 0$.

1.3.2 Geometrical Interpretation of the Strain Components

In general $\mathrm{d}s^2 - \mathrm{d}S^2 = 2E_{KL}\,\mathrm{d}X_K\,\mathrm{d}X_L$, which in the particular case where $\mathrm{d}\boldsymbol{X} = (\mathrm{d}X_1, 0, 0)$ gives

$$\mathrm{d}s^2 - \mathrm{d}X_1^2 = 2E_{11}\,\mathrm{d}X_1^2 \;\Rightarrow\; \mathrm{d}s^2 = (1+2E_{11})\,\mathrm{d}X_1^2,$$

so the line element $\mathrm{d}X_1$ is stretched in the ratio $(1+2E_{11})^{1/2}$.

For small strains $|E_{11}| \ll 1$ we have

$$(1+2E_{11})^{1/2} \sim 1 + E_{11} + O(E_{11}^2), \qquad \text{Binomial Theorem}$$
$$\Rightarrow\; \mathrm{d}s \approx (1+E_{11})\,\mathrm{d}X_1 \;\Rightarrow E_{11} \approx \frac{\mathrm{d}s - \mathrm{d}X_1}{\mathrm{d}X_1}.$$

This is roughly extension divided by original length. Similar comments apply to E_{22} and E_{33}. E_{11}, E_{22}, E_{33} are called Normal Strains.

The Lagrangian strain tensor is a symmetric tensor

$$\begin{aligned} 2E_{KL} &= \frac{\partial x_k}{\partial X_K}\frac{\partial x_k}{\partial X_L} - \delta_{KL}, \\ &= \frac{\partial x_k}{\partial X_L}\frac{\partial x_k}{\partial X_K} - \delta_{LK} = 2E_{LK}. \end{aligned}$$

Example 1.3.7. *Consider the deformation of a rectangular parallelepiped with no deformation along the X_3 axis. A parallelepiped is a prism whose bases are parallelograms.*

Our objective is to obtain an interpretation of the off-diagonal components of the Lagrangian strain tensor in particular the component E_{12} in terms of θ_{12}.

θ_{12} is the deviation from perpendicular of the two sides initially of length $\mathrm{d}X_1$ and $\mathrm{d}X_2$.

Let $\mathrm{d}S$ ($\mathrm{d}s$) be the length of the diagonal before (after) deformation so [applying the cosine rule]

$$\mathrm{d}s^2 = (1+2E_{11})\,\mathrm{d}X_1^2 + (1+2E_{22})\,\mathrm{d}X_2^2 - 2\sqrt{1+2E_{11}}\sqrt{1+2E_{22}}\,\mathrm{d}X_1\,\mathrm{d}X_2\cos(\pi/2+\theta_{12}).$$

But $\mathrm{d}s^2 - \mathrm{d}S^2 = 2E_{KL}\,\mathrm{d}X_K\,\mathrm{d}X_L$ so that

$$\begin{aligned} \mathrm{d}s^2 &= \mathrm{d}X_1^2 + \mathrm{d}X_2^2 + 2E_{11}\,\mathrm{d}X_1^2 + 2E_{22}\,\mathrm{d}X_2^2 + 4E_{12}\,\mathrm{d}X_1\,\mathrm{d}X_2, \\ &= (1+2E_{11})\,\mathrm{d}X_1^2 + (1+2E_{22})\,\mathrm{d}X_2^2 + 4E_{12}\,\mathrm{d}X_1\,\mathrm{d}X_2. \end{aligned}$$

So equating these two equations for $\mathrm{d}s^2$ *gives*

$$\begin{aligned} -2\sqrt{1+2E_{11}}\sqrt{1+2E_{22}}\,\mathrm{d}X_1\,\mathrm{d}X_2\cos(\pi/2+\theta_{12}) &= 4E_{12}\,\mathrm{d}X_1\,\mathrm{d}X_2, \\ \sqrt{1+2E_{11}}\sqrt{1+2E_{22}}\sin(\theta_{12}) &= 2E_{12}, \\ \Rightarrow \sin(\theta_{12}) &= \frac{2E_{12}}{\sqrt{1+2E_{11}}\sqrt{1+2E_{22}}}. \end{aligned}$$

The effect of the deformation is to transform the initial rectangular parallelepiped into a non-rectangular parallelepiped.

For small deformation $|E_{kl}| \ll 1$ *implies that* $\theta_{12} \approx 2E_{12}$*. This implies that* E_{12} *represent half the angle of rotation. In comparison, with the diagonal elements which are associated with change of volume* E_{12} *is associated with a change of shape.* E_{12}, E_{13}, E_{23} *are called Shear Strains.*

Example 1.3.8. *Consider the deformation* $x_1 = \lambda X_1$*,* $x_2 = \lambda X_2$*,* $x_3 = \lambda X_3$*, and so the Lagrangian Strain Tensor is*

$$\begin{aligned} 2E_{11} &= x_{k,1}x_{k,1} - 1 = \lambda^2 - 1, \\ 2E_{22} &= \lambda^2 - 1, \\ 2E_{33} &= \lambda^2 - 1, \end{aligned}$$

$$E_{12} = E_{13} = E_{23} = 0.$$

The body is extended (compressed if $\lambda < 1$*) equally in all directions with angles preserved. This change of volume with no change of shape is called* pure dilatation.

Definition 1.3.9. $E_{kk} = E_{11} + E_{22} + E_{33}$ *is called the Dilatation. It is often denoted by* Δ*.*

1.3.3 Strain in terms of displacement

We now express E_{KL} in terms of the displacement gradient. We define the displacement gradient $\boldsymbol{U} = (U_1, U_2, U_3)^T$ to be

$$\boldsymbol{U}(\boldsymbol{X}, t) = \boldsymbol{x} - \boldsymbol{X}.$$

So, using the summation convention this can be written as $x_m = X_m + U_m(\boldsymbol{X}, t)$, differentiating this with respect to X_K gives

$$\frac{\partial x_m}{\partial X_K} = \frac{\partial X_m}{\partial X_K} + \frac{\partial U_m}{\partial X_K} = \delta_{mK} + \frac{\partial U_m}{\partial X_K}.$$

Substituting into the definition of the Lagrangian Strain Tensor E_{KL} we have

$$\begin{aligned} E_{KL} &= \frac{1}{2}\left\{\left(\delta_{mK} + \frac{\partial U_m}{\partial X_K}\right)\left(\delta_{mL} + \frac{\partial U_m}{\partial X_L}\right) - \delta_{KL}\right\}, \\ &= \frac{1}{2}\left[\delta_{mK}\delta_{mL} + \delta_{mK}\frac{\partial U_m}{\partial X_L} + \delta_{mL}\frac{\partial U_m}{\partial X_K} + \frac{\partial U_m}{\partial X_K}\frac{\partial U_m}{\partial X_L} - \delta_{KL}\right], \\ &= \frac{1}{2}\left[\delta_{KL} + \frac{\partial U_K}{\partial X_L} + \frac{\partial U_L}{\partial X_K} + \frac{\partial U_m}{\partial X_K}\frac{\partial U_m}{\partial X_L} - \delta_{KL}\right], \\ &= \frac{1}{2}\left[\frac{\partial U_K}{\partial X_L} + \frac{\partial U_L}{\partial X_K} + \frac{\partial U_m}{\partial X_K}\frac{\partial U_m}{\partial X_L}\right]. \end{aligned}$$

Definition 1.3.10. *The linearized form of the Lagrangian Strain Tensor is suitable for small displacement gradients and is given by*

$$\boxed{E_{KL} = \frac{1}{2}\left[\frac{\partial U_K}{\partial X_L} + \frac{\partial U_L}{\partial X_K}\right]}$$

We will mainly be concerned with small displacements in linear elasticity.

Example 1.3.11. *For small displacements, we have*

$$E_{11} = \frac{1}{2}\left[\frac{\partial U_1}{\partial X_1} + \frac{\partial U_1}{\partial X_1}\right] = \frac{\partial U_1}{\partial X_1} \qquad E_{12} = \frac{1}{2}\left[\frac{\partial U_1}{\partial X_2} + \frac{\partial U_2}{\partial X_1}\right]$$
$$E_{22} = \frac{1}{2}\left[\frac{\partial U_2}{\partial X_2} + \frac{\partial U_2}{\partial X_2}\right] = \frac{\partial U_2}{\partial X_2} \qquad E_{13} = \frac{1}{2}\left[\frac{\partial U_1}{\partial X_3} + \frac{\partial U_3}{\partial X_1}\right]$$

Definition 1.3.12 (Rotation Tensor)**.** *Notice that we can also write*

$$\begin{aligned}\frac{\partial U_K}{\partial X_L} &= \frac{1}{2}\left(\frac{\partial U_K}{\partial X_L} + \frac{\partial U_L}{\partial X_K}\right) + \frac{1}{2}\left(\frac{\partial U_K}{\partial X_L} - \frac{\partial U_L}{\partial X_K}\right)\\ &= E_{KL} + R_{KL}\end{aligned}$$

Where E_{KL} is the Extension and Shear, and R_{KL} is the rotation. Also note that unlike E_{KL}, R_{KL} is an antisymmetric tensor.

Example 1.3.13 (Simple Shear)**.** *$U_1 = 2\varepsilon X_2$, and $U_2 = U_3 = 0$, for $\varepsilon > 0$ and $\varepsilon \ll 1$, so we have*

$$\frac{\partial U_1}{\partial X_2} = 2\varepsilon \quad \textit{and otherwise} \quad \frac{\partial U_K}{\partial X_L} = 0$$

So $E_{12} = E_{21} = \frac{1}{2}(2\varepsilon + 0) = \varepsilon$, are the only non-zero values of E_{KL}. Also $R_{12} = \varepsilon$ and $R_{21} = -\varepsilon$, otherwise $R_{KL} = 0$. In this case we have shear and rotation but no extension.

Example 1.3.14 (Pure Shear)**.** *$U_1 = \varepsilon X_2$, $U_2 = \varepsilon X_1$, and $U_3 = 0$, for $\varepsilon > 0$ and $\varepsilon \ll 1$.*
It is easy to show that $E_{12} = E_{21} = \varepsilon$ and all other strain tensors are zero All the R_{KL}'s are zero.
This represents no extension or rotation , just shear - as the name would suggest.

1.3.4 Principal Axes

For any second order symmetrical cartesian tensor it is always possible to choose axes, called principal axes, such that the matrix of components of the tensor is of diagonal form.

Theorem 1.3.15. *If $\boldsymbol{e}^{(1)}$,$\boldsymbol{e}^{(2)}$ and $\boldsymbol{e}^{(3)}$ are eigenvectors of the second order symmetrical Cartesian tensor A, with corresponding eigenvalues $\lambda_1, \lambda_2, \lambda_3$ then relative to the axes given by eigenvectors the components of A are $a_{11} = \lambda_1$, $a_{22} = \lambda_2$ and $a_{33} = \lambda_3$ and $a_{rs} = 0$ otherwise. The eigenvectors are the principal axes.*

Example 1.3.16. *Find the principal axes of the Lagrangian Strain tensor whose matrix of components if*

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

The characteristic equation is $-\lambda[(1-\lambda)^2 - 1] = \lambda^2(\lambda - 2) = 0$, thus $\lambda_1 = 2$ $\lambda_2 = \lambda_3 = 0$ and . Recall from your linear algebra[1] *that the eigenvector $(v_1, v_2, v_3)^T$ is such that*

$$\begin{aligned} -\lambda v_1 &= 0 \\ (1-\lambda)v_2 + v_3 &= 0 \\ v_2 + (1-\lambda)v_3 &= 0. \end{aligned}$$

so we conclude that $\lambda = 2$ have eigenvectors $\boldsymbol{e}^{(1)} = (0,1,1)^T/\sqrt{2}$ and $\lambda = 0$ has eigenvectors $\boldsymbol{e}^{(2)} = (1,0,0)^T$ and $\boldsymbol{e}^{(3)} = (0,1,-1)^T/\sqrt{2}$.
If $\boldsymbol{e}^{(1)}$,$\boldsymbol{e}^{(2)}$ and $\boldsymbol{e}^{(3)}$ are the principal axes then in this coordinate system, the matrix of the strain tensor becomes

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Example 1.3.17 (Pure Shear)**.** *$x_1 = X_1 + \varepsilon X_2$, $x_2 = X_2 + \varepsilon X_1$ and $x_3 = X_3$, for $\varepsilon > 0$ and finite, thus we include the non-linear terms since we have finite displacement, thus it can be shown that*

$$E_{KL} = \begin{pmatrix} \varepsilon^2/2 & \varepsilon & 0 \\ \varepsilon & \varepsilon^2/2 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

which has eigenvalues $\lambda = \varepsilon + \varepsilon^2/2, -\varepsilon + \varepsilon^2/2, 0$ and associated eigenvectors $\boldsymbol{e}^{(1)} = \frac{1}{\sqrt{2}}(1,1,0)^T$ and $\boldsymbol{e}^{(2)} = \frac{1}{\sqrt{2}}(-1,1,0)^T$ and $\boldsymbol{e}^{(3)} = (0,0,1)^T$.

[1] An Eigenvector $\boldsymbol{x}$, with eigenvalue λ, will be a solution to $A\boldsymbol{x} = \lambda\boldsymbol{x}$

If we perform the anticlockwise rotation by $\pi/4$ the principal axes

$$\begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X_1' \\ X_2' \\ X_3' \end{pmatrix}.$$

Thus we obtain

$$\mathrm{d}s^2 - \mathrm{d}S^2 = 2(\varepsilon + \varepsilon^2/2)\,\mathrm{d}X_1'\,\mathrm{d}X_1' + 2(-\varepsilon + \varepsilon'2)\,\mathrm{d}X_2'\,\mathrm{d}X_2'.$$

Compare this with the figure for pure shear.

1.3.5 Invariants of second order symmetric tensor

Example 1.3.18. *Show that the eigenvalues of a second order symmetric tensor are independent of the coordinate system.*

Suppose λ is an eigenvalue of σ_{rs}' with respect to $\boldsymbol{i}_1', \boldsymbol{i}_2'$ and $\boldsymbol{i}_3'$, then

$$\sigma_{rs}' x_s' = \lambda x_r'.$$

Using the definition of a second order tensor $\sigma_{rs}' = l_{ri} l s j \sigma_{ij}$ and also $x_s' = l_{sp} x_p$ and $x_r' = l_{rq} x_q$, so we have

$$l_{ri}\sigma_{ij} l_{sj} l_{sp} x_p = \lambda l_{rq} x_q.$$

Multiplying through by l_{rm} yield

$$\begin{aligned} l_{rm} l_{ri} \sigma_{ij} l_{sj} l_{sp} x_p &= \lambda l_{rm} l_{rq} x_q, \\ \delta_{mi} \sigma_{ij} \delta_{jp} x_p &= \lambda \delta_{mq} x_q, \\ \sigma_{mj} x_j &= \lambda x_m. \end{aligned}$$

so λ is an eigenvalue of σ_{mj} with respect to $\boldsymbol{i}_1, \boldsymbol{i}_2$ and $\boldsymbol{i}_3$

The eigenvalues λ_1, λ_2 and λ_3 of a second order symmetric tensor are therefore said to invariants of σ_{ij}. Any function $f(\lambda_1, \lambda_2, \lambda_3)$ who's values is unaltered by interchanges of pair of $\lambda_1, \lambda_2, \lambda_3$ is also an invariant of the tensor σ_{ij}.

1.3.6 Compatibility Conditions

The components of the linear strain tensor in terms of the displacement are

$$E_{KL} = \frac{1}{2}\left(\frac{\partial U_K}{\partial X_L} + \frac{\partial U_L}{\partial X_K}\right)$$

If displacements U_1, U_2 and U_3 are prescribed continuous functions of X_1, X_2 and X_3, then the strain components can be uniquely determined.
On the other hand, if the strain components are prescribed function of the coordinates, it will not be possible to find unique values for the displacements because strain represents pure deformation whereas the displacements include both deformation and rigid body motion. The strain displacement relations form a system of six partial differential equations with only three unknowns U_1, U_2 and U_3, it is obvious that some restrictions must be imposed on the strains. The compatibility condition are necessary conditions for the strain components to give single value displacements. (N.B. If strain are obtained from displacements, then there is no need to check compatibility conditions.) The compatibility conditions are given by:

$$\begin{aligned}
\frac{\partial^2 E_{11}}{\partial X_2 \partial X_3} &= \frac{\partial}{\partial X_1}\left(-\frac{\partial E_{23}}{\partial X_1} + \frac{\partial E_{31}}{\partial X_2} + \frac{\partial E_{12}}{\partial X_3}\right),\\
\frac{\partial^2 E_{22}}{\partial X_3 \partial X_1} &= \frac{\partial}{\partial X_2}\left(-\frac{\partial E_{31}}{\partial X_2} + \frac{\partial E_{12}}{\partial X_3} + \frac{\partial E_{23}}{\partial X_1}\right),\\
\frac{\partial^2 E_{33}}{\partial X_1 \partial X_2} &= \frac{\partial}{\partial X_3}\left(-\frac{\partial E_{12}}{\partial X_3} + \frac{\partial E_{23}}{\partial X_1} + \frac{\partial E_{31}}{\partial X_2}\right),\\
2\frac{\partial^2 E_{12}}{\partial X_1 \partial X_2} &= \frac{\partial^2 E_{11}}{\partial X_2^2} + \frac{\partial^2 E_{22}}{\partial X_1^2},\\
2\frac{\partial^2 E_{23}}{\partial X_2 \partial X_3} &= \frac{\partial^2 E_{22}}{\partial X_3^2} + \frac{\partial^2 E_{33}}{\partial X_2^2},\\
2\frac{\partial^2 E_{31}}{\partial X_3 \partial X_1} &= \frac{\partial^2 E_{33}}{\partial X_1^2} + \frac{\partial^2 E_{11}}{\partial X_3^2}.
\end{aligned}$$

Example 1.3.19. *We shall proceed to check the first compatibility condition, we have*

$$E_{11} = \frac{\mathrm{d}U_1}{\mathrm{d}X_1}; \quad 2E_{23} = \frac{\partial U_2}{\partial X_3} + \frac{\partial U_3}{\partial X_2}; \quad 2E_{31} = \frac{\partial U_3}{\partial X_1} + \frac{\partial U_1}{\partial X_3}; \quad 2E_{12} = \frac{\partial U_1}{\partial X_2} + \frac{\partial U_2}{\partial X_1}$$

Substituting this into the RHS of the equation yields

$$\begin{aligned}
&\frac{\partial}{\partial X_1}\left(-\frac{\partial E_{23}}{\partial X_1} + \frac{\partial E_{31}}{\partial X_2} + \frac{\partial E_{12}}{\partial X_3}\right),\\
=&\frac{1}{2}\frac{\partial}{\partial X_1}\left(-\frac{\partial^2 U_2}{\partial X_1 \partial X_3} - \frac{\partial^2 U_3}{\partial X_1 \partial X_2} + \frac{\partial^2 U_3}{\partial X_1 \partial X_2} + \frac{\partial^2 U_1}{\partial X_2 \partial X_3} + \frac{\partial^2 U_1}{\partial X_2 \partial X_3} + \frac{\partial^2 U_2}{\partial X_1 \partial X_3}\right),\\
=&\frac{\partial}{\partial X_1}\frac{\partial^2 U_1}{\partial X_2 \partial X_3},\\
=&\frac{\partial^2}{\partial X_2 \partial X_3}\left(\frac{\partial U_1}{\partial X_1}\right),\\
=&\frac{\partial^2 E_{11}}{\partial X_2 \partial X_3},
\end{aligned}$$

So the first compatibility condition is a necessary condition

Example 1.3.20 (Uniaxial Tension - Examples 3, Q3)**.** *Consider the Lagrangian Strain Tensor given by $E_{11} = A$, $E_{22} = E_{33} = B$ and all others zero, where A and B are constants.*
All the compatibility conditions are satisfied, and the displacements satisfy the linearized form of Lagrangian Strain Tensor, namely

$$\frac{\partial U_1}{\partial X_1} = A, \quad \frac{\partial U_2}{\partial X_2} = \frac{\partial U_3}{\partial X_3} = B, \quad \frac{\partial U_1}{\partial X_2} + \frac{\partial U_2}{\partial X_1} = 0,$$

$$\frac{\partial U_2}{\partial X_3} + \frac{\partial U_3}{\partial X_2} = 0 \quad \textit{and} \quad \frac{\partial U_1}{\partial X_3} + \frac{\partial U_3}{\partial X_1} = 0.$$

We can now solve for U_1, U_2 and U_3. It is not possible to find unique values for U_1, U_2, U_3 and in general they include constants of integration which represent rigid body motion.

1.3.7 Rate of Change of a vector

In Lagrangian coordinates $\boldsymbol{F}(\boldsymbol{X}, t) = F_K(\boldsymbol{X}, t)\boldsymbol{i}_K$, then since $\partial X_K/\partial t = 0$;

$$\frac{\mathrm{d}\boldsymbol{F}}{\mathrm{d}t} = \frac{\partial F_K}{\partial t}\boldsymbol{i}_K.$$

In Eulerian coordinates $\boldsymbol{f}(\boldsymbol{x}, t) = f_k(\boldsymbol{x}, t)\boldsymbol{i}_k$, then by the chain rule

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{f} = \left(\frac{\partial f_k}{\partial t} + \left(\frac{\partial x_l}{\partial t}\right)_{\boldsymbol{X}} \frac{\partial f_k}{\partial x_l}\right)\boldsymbol{i}_k.$$

The velocity of a particle is $\boldsymbol{v} = v_k\boldsymbol{i}_k = \left(\dfrac{\partial x_k}{\partial t}\right)_{\boldsymbol{X}} \boldsymbol{i}_k$

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{f} = \left(\frac{\partial f_k}{\partial t} + v_l\frac{\mathrm{d}f_k}{\mathrm{d}x_l}\right)\boldsymbol{i}_k = \left(\frac{\mathrm{D}f_k}{\mathrm{D}t}\right)\boldsymbol{i}_k = \frac{\mathrm{D}\boldsymbol{f}}{\mathrm{D}t}.$$

The operator $\dfrac{\mathrm{D}}{\mathrm{D}t} = \dfrac{\partial}{\partial t} + v_l\dfrac{\partial}{\partial x_l}$ is known as the *Material Derivative.*

1.3.8 Velocity and Acceleration

A. Elasticity (Lagrangian Coordinates)

- Displacement $\boldsymbol{U}(\boldsymbol{X}, t) = U_K(\boldsymbol{X}, t)\boldsymbol{i}_K$.
- Velocity $\boldsymbol{V} = \dfrac{\mathrm{d}\boldsymbol{U}}{\mathrm{d}t} = \dfrac{\partial U_K}{\partial t}\boldsymbol{i}_K = V_K\boldsymbol{i}_K$

- Acceleration $\boldsymbol{A} = \dfrac{\mathrm{d}\boldsymbol{V}}{\mathrm{d}t} = \dfrac{\partial}{\partial t} V_K \boldsymbol{i}_K = \dfrac{\partial^2 U_K}{\partial t^2} \boldsymbol{i}_K$

B. Fluid Mechanics (Eulerian coordinates)

- Velocity $\boldsymbol{v} = \boldsymbol{v}(\boldsymbol{x}, t) = v_k(\boldsymbol{x}, t)\boldsymbol{i}_k$

- Acceleration $\boldsymbol{a} = \dfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{v} = \dfrac{\mathrm{D}}{\mathrm{D}t}\boldsymbol{v}$

1.3.9 The Rate of Strain Tensor

We recall that $x_k = x_k(\boldsymbol{X}, t)$ is the Lagrangian description of motion. Consider the rate-of-change of an element

$$\mathrm{d}x_k = x_{k,K}\, \mathrm{d}X_K = \frac{\partial x_k}{\partial X_K}\, \mathrm{d}X_K.$$

Differentiating this with respect to t

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}(\ \mathrm{d}x_k) &= \frac{\mathrm{d}}{\mathrm{d}t}(x_{k,K}\, \mathrm{d}X_K), \\
&= \frac{\mathrm{d}}{\mathrm{d}t}(x_{k,K})\, \mathrm{d}X_K, && \mathrm{d}X_K \text{ is Lagrangian} \\
&= \frac{\partial}{\partial X_K}\left(\frac{\partial x_k}{\partial t}\right) \mathrm{d}X_K, \\
&= \frac{\partial v_k}{\partial X_K}\, \mathrm{d}X_K, \\
&= \frac{\partial v_k}{\partial x_l}\frac{\partial x_l}{\partial X_K}\, \mathrm{d}X_K, && \text{Chain Rule} \\
&= \frac{\partial v_k}{\partial x_l}\ \mathrm{d}x_l.
\end{aligned}$$

Example 1.3.21. *Show that (a)* $x_{l,L}X_{L,k} = \delta_{kl}$ *and (b)* $\dfrac{\mathrm{d}}{\mathrm{d}t}(X_{K,k}) = -X_{K,l}v_{l,k}$.

1.3.10 Deformation rate tensor

We now derive a formula for the deformation rate tensor by considering the rate of change of the square of length $\mathrm{d}\boldsymbol{x}$:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}(\,\mathrm{d}s^2) &= \frac{\mathrm{d}}{\mathrm{d}t}\left(\,\mathrm{d}x_k\,\mathrm{d}x_k\right), \\
&= 2\frac{\mathrm{d}}{\mathrm{d}t}(\,\mathrm{d}x_k)\,\mathrm{d}x_k, \\
&= 2\frac{\partial v_k}{\partial x_l}\;\mathrm{d}x_l\;\mathrm{d}x_k, \\
&= 2\frac{\partial v_l}{\partial x_k}\;\mathrm{d}x_k\;\mathrm{d}x_l, && \text{Rate of Strain} \\
&= \left(\frac{\partial v_k}{\partial x_l}+\frac{\partial v_l}{\partial x_k}\right)\;\mathrm{d}x_l\;\mathrm{d}x_k, \\
&= 2d_{kl}\;\mathrm{d}x_k\;\mathrm{d}x_l.
\end{aligned}$$

Definition 1.3.22. *We define the deformation rate tensor d_{kl} by*

$$\mathrm{d}_{kl} = \frac{1}{2}\left(\frac{\partial v_k}{\partial x_l}+\frac{\partial v_l}{\partial x_k}\right).$$

We recall that $\mathrm{d}s^2 - \mathrm{d}S^2 = 2E_{KL}\,\mathrm{d}X_K\,\mathrm{d}X_L$. We now differentiate this with respect to time:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}(\,\mathrm{d}s^2 - \mathrm{d}S^2) = \frac{\mathrm{d}}{\mathrm{d}t}(\,\mathrm{d}s^2) &= \frac{\mathrm{d}}{\mathrm{d}t}(2E_{KL}\,\mathrm{d}X_K\,\mathrm{d}X_L), \\
&= 2\frac{\mathrm{d}E_{KL}}{\mathrm{d}t}\,\mathrm{d}X_K\,\mathrm{d}X_L, \\
&= 2\dot{E}_{KL}X_{K,k}X_{L,l}\;\mathrm{d}x_k\;\mathrm{d}x_l.
\end{aligned}$$

We deduce by equating to the above statement

$$d_{kl} = \dot{E}_{KL}X_{K,k}X_{L,l}.$$

We recall also that $\mathrm{d}s^2 - \mathrm{d}S^2 = 2e_{kl}\;\mathrm{d}x_k\;\mathrm{d}x_l$ and that $\mathrm{d}s^2 - \mathrm{d}S^2 = 2E_{KL}\,\mathrm{d}X_K\,\mathrm{d}X_L = 2E_{KL}X_{K,k}X_{L,l}\;\mathrm{d}x_k\;\mathrm{d}x_l$ therefore $e_{kl} = E_{KL}X_{K,k}X_{L,l}$. Differentiating this expression yields

$$\begin{aligned}
\dot{e}_{kl} &= \dot{E}_{KL}X_{K,k}X_{L,l} + E_{KL}\dot{X}_{K,k}X_{L,l} + E_{KL}X_{K,k}\dot{X}_{L,l}, \\
&= d_{kl} - E_{KL}X_{K,m}v_{m,k}X_{L,l} - E_{KL}X_{K,k}X_{L,n}v_{n,l}, \\
&= d_{kl} - e_{ml}v_{m,k} - e_{kn}v_{n,l}.
\end{aligned}$$

Having used exercise b) above and the above results, thus we have

$$\boxed{d_{kl} = \dot{e}_{kl} + e_{ml}v_{m,k} + e_{kn}v_{n,l}.}$$

or for small deformations

$$d_{kl} = \dot{e}_{kl}.$$

1.4 Stress

1.4.1 The Stress Tensor σ_{ij}

A body undergoes deformation when subjected to forces. These forces may be mechanical, electrical etc. There are three types of forces acting on a volume Ω contained within a surface $\partial\Omega$

1. Internal forces between pairs of particles belonging to Ω for example electrical.
2. External forces acting on the material within $\partial\Omega$, known as body forces for example gravity.
3. Surface forces, transmitted across the boundary of Ω from the region outside of Ω

We defined σ_{ij} to be the Cartesian stress tensor. The first suffix corresponds to the direction of the normal to the surface on which the stress acts. The second suffix refers to the component of stress.

Definition 1.4.1. *If the unit normal to the boundary of a volume is given by* $\boldsymbol{n} = (n_1, n_2, n_3)^T$ *then the stress component normal to the surface are given by* $T_j = \sigma_{ij} n_i$. *Here,* $\boldsymbol{T}$ *is the stress vector.*

Example 1.4.2. *1.* $\boldsymbol{n} = (1, 0, 0)^T$, *thus*
$T_1 = \sigma_{11} n_1 + \sigma_{21} n_2 + \sigma_{31} n_3 = \sigma_{11}$,
$T_2 = \sigma_{12}$,
$T_3 = \sigma_{13}$.
Stress vector across the surface with unit normal $\boldsymbol{n}$ *is* $\boldsymbol{T} = (\sigma_{11}, \sigma_{12}, \sigma_{13})^T$.

2. $\boldsymbol{n} = (-1, 0, 0)^T$, *thus*
$T_1 = -\sigma_{11}$,
$T_2 = -\sigma_{12}$,
$T_3 = -\sigma_{13}$.
Stress vectors across the surface with unit normal $\boldsymbol{n}$ *is* $\boldsymbol{T} = -(\sigma_{11}, \sigma_{12}, \sigma_{13})^T$.

3. $\boldsymbol{n} = (0,1,0)^T$, *thus*
   $T_1 = \sigma_{21}$,
   $T_2 = \sigma_{22}$,
   $T_3 = \sigma_{23}$,
   $\Rightarrow\ \boldsymbol{T} = (\sigma_{21}, \sigma_{22}, \sigma_{23})^T$.

1.4.2 Properties of σ_{ij}

1. Show that σ_{ij} is a tensor.
   Let $L = [l_{ij}]$, and a rotation with $\boldsymbol{x'} = L\boldsymbol{x}$, we write the Stress vectors $\boldsymbol{T} = S\boldsymbol{n}$ with S representing the matrix of the stress tensor. So that $\boldsymbol{T}$ and $\boldsymbol{n}$ will transforms to $\boldsymbol{T'} = L\boldsymbol{T}$ and $\boldsymbol{n'} = L\boldsymbol{n}$, so that in the transformed coordinate system S' satisfies

$$\begin{aligned}
& & \boldsymbol{T'} &= S'\boldsymbol{n'}, & \\
\Rightarrow & & L\boldsymbol{T} &= S'L\boldsymbol{n}, & \\
\Rightarrow & & LS\boldsymbol{n} &= S'L\boldsymbol{n}, & \\
\Rightarrow & & LSL^T &= S'LL^T = S', & \text{orthogonality of } L \\
\Rightarrow & & \sigma'_{ij} &= l_{i\alpha}l_{j\beta}\sigma_{\alpha\beta}. &
\end{aligned}$$

   Thus σ_{ij} is a second order cartesian tensor.

2. Show that σ_{ij} is symmetric
   We consider for simplicity, a rectangular element in two dimensions and apply conservation of angular momentum.

   Rate of change of angular momentum about the centre of gravity G is

$$2(\sigma_{12}\delta x_2)\frac{\delta x_1}{2} - 2(\sigma_{21}\delta x_1)\frac{\delta x_2}{2} = 0.$$

   Letting the rectangle shrink to zero and assume that angular acceleration remains finite yields $\sigma_{12} = \sigma_{21}$. More generally it can be shown that $\sigma_{ij} = \sigma_{ji}$.

1.5 The Transport Theorem

Example 1.5.1. *The Jacobian of the transformation between Eulerian* (x_1, x_2, x_3) *to Lagrangian coordinates* (X_1, X_2, X_3) *defined by*

$$J = \frac{\partial(x_1, x_2, x_3)}{\partial(X_1, X_2, X_3)}$$

satisfies the equation $\dfrac{\mathrm{d}J}{\mathrm{d}t} = J\mathrm{div}(\boldsymbol{v})$.

Theorem 1.5.2 (The Transport Theorem)**.** *Consider a material volume $V(t)$, that is a volume consisting of the same material particles at any time then*

$$\frac{\mathrm{d}}{\mathrm{d}t}\left\{\int_{V(t)} L(\boldsymbol{x},t)\ \mathrm{d}x_1\ \mathrm{d}x_2\ \mathrm{d}x_3\right\} = \int_{V(t)}\left\{\frac{\mathrm{d}L}{\mathrm{d}t} + L\mathrm{div}(\boldsymbol{v})\right\}\ \mathrm{d}x_1\ \mathrm{d}x_2\ \mathrm{d}x_3.$$

This is a formula for "differentiating" over a volume which is moving with the fluid.

Proof.

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\left\{\int_{V(t)} L(\boldsymbol{x},t)\ \mathrm{d}x_1\ \mathrm{d}x_2\ \mathrm{d}x_3\right\} &= \frac{\mathrm{d}}{\mathrm{d}t}\left\{\int_{V(0)} L(\boldsymbol{x}(\boldsymbol{X},t),t)J\ \mathrm{d}X_1\,\mathrm{d}X_2\,\mathrm{d}X_3\right\},\\
&= \int_{V(0)}\frac{\mathrm{d}}{\mathrm{d}t}\left\{L(\boldsymbol{x}(\boldsymbol{X},t),t)J\right\}\ \mathrm{d}X_1\,\mathrm{d}X_2\,\mathrm{d}X_3,\\
&= \int_{V(0)}\left\{\frac{\mathrm{d}L}{\mathrm{d}t}J + L\frac{\mathrm{d}J}{\mathrm{d}t}\right\}\ \mathrm{d}X_1\,\mathrm{d}X_2\,\mathrm{d}X_3,\\
&= \int_{V(0)}\left\{\frac{\mathrm{d}L}{\mathrm{d}t}J + LJ\mathrm{div}(\boldsymbol{v})\right\}\ \mathrm{d}X_1\,\mathrm{d}X_2\,\mathrm{d}X_3,\\
&= \int_{V(0)}\left\{\frac{\mathrm{d}L}{\mathrm{d}t} + L\mathrm{div}(\boldsymbol{v})\right\}J\ \mathrm{d}X_1\,\mathrm{d}X_2\,\mathrm{d}X_3,\\
&= \int_{V(t)}\left\{\frac{\mathrm{d}L}{\mathrm{d}t} + L\mathrm{div}(\boldsymbol{v})\right\}\ \mathrm{d}x_1\ \mathrm{d}x_2\ \mathrm{d}x_3.
\end{aligned}$$

□

1.5.1 Conservation of Mass

Let $\rho(\boldsymbol{x},t)$ be the fluid density, then the conservation of mass principle is given by

$$\frac{\mathrm{d}}{\mathrm{d}t}\left\{\int_{V(t)} \rho\ \mathrm{d}x_1\ \mathrm{d}x_2\ \mathrm{d}x_3\right\} = 0.$$

Applying the transport theorem yields

$$\int_{V(t)}\left\{\frac{\mathrm{d}\rho}{\mathrm{d}t} + \rho\mathrm{div}(\boldsymbol{v})\right\}\ \mathrm{d}x_1\ \mathrm{d}x_2\ \mathrm{d}x_3 = 0.$$

since $V(t)$ is arbitrary the integrand is equal to zero, namely $\dfrac{\mathrm{d}\rho}{\mathrm{d}t} + \rho\mathrm{div}(\boldsymbol{v}) = 0$.

$$\begin{aligned}
\frac{\mathrm{d}\rho}{\mathrm{d}t} &= \frac{\partial\rho}{\partial t} + \frac{\partial\rho}{\partial x_1}\frac{\partial x_1}{\partial t} + \frac{\partial\rho}{\partial x_2}\frac{\partial x_2}{\partial t} + \frac{\partial\rho}{\partial x_3}\frac{\partial x_3}{\partial t},\\
&= \frac{\partial\rho}{\partial t} + \boldsymbol{v}.\nabla\rho,\\
&= \frac{\mathrm{D}\rho}{\mathrm{D}t}.
\end{aligned}$$

Thus we have that

$$\boxed{\frac{\mathrm{D}\rho}{\mathrm{D}t} + \rho \mathrm{div}(\boldsymbol{v}) = 0 \quad \text{or} \quad \frac{\partial \rho}{\partial t} + v_l \frac{\partial \rho}{\partial x_l} + \rho \frac{\partial v_k}{\partial x_k} = 0.}$$

If the density ρ is constant, as we may experience with say water then $\dfrac{\partial \rho}{\partial t} = \dfrac{\partial \rho}{\partial x_l} = 0$, then the conservation of mass reduces to

$$\boxed{\nabla \cdot (\boldsymbol{v}) = 0 \qquad \text{or} \qquad \frac{\partial v_k}{\partial x_k} = 0.}$$

1.5.2 Conservation of Momentum

By the transport theorem we consider the conservation of momentum.

$$\frac{\mathrm{d}}{\mathrm{d}t}\left\{\int_{V(t)} \rho v_i \ \mathrm{d}x_1 \ \mathrm{d}x_2 \ \mathrm{d}x_3\right\} = \int_{V(t)} \left\{\frac{\mathrm{d}}{\mathrm{d}t}(\rho v_i) + \rho v_i \frac{\partial v_k}{\partial x_k}\right\} \ \mathrm{d}x_1 \ \mathrm{d}x_2 \ \mathrm{d}x_3.$$

Take

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}(\rho v_i) + \rho v_i \frac{\partial v_k}{\partial x_k}, &= \frac{\partial}{\partial t}(\rho v_i) + v_m \frac{\partial}{\partial x_m}(\rho v_i) + \rho v_i \frac{\partial v_k}{\partial x_k}, \\
&= \rho \frac{\partial v_i}{\partial t} + v_i \frac{\partial \rho}{\partial t} + \rho v_m \frac{\partial v_i}{\partial x_m} + v_i v_m \frac{\partial \rho}{\partial x_m} + \rho v_i \frac{\partial v_k}{\partial x_k}, \\
&= \rho \frac{\partial v_i}{\partial t} + \rho v_m \frac{\partial v_i}{\partial x_m} + v_i \left(\frac{\partial \rho}{\partial t} + v_m \frac{\partial \rho}{\partial x_m} + \rho \frac{\partial v_k}{\partial x_k}\right), \\
&= \rho \frac{\partial v_i}{\partial t} + \rho v_m \frac{\partial v_i}{\partial x_m}, \\
&= \rho \frac{\mathrm{D}v_i}{\mathrm{D}t}.
\end{aligned}$$

Hence

$$\frac{\mathrm{d}}{\mathrm{d}t}\left\{\int_{V(t)} \rho v_i \ \mathrm{d}x_1 \ \mathrm{d}x_2 \ \mathrm{d}x_3\right\} = \int_{V(t)} \rho \frac{\mathrm{D}v_i}{\mathrm{D}t} \ \mathrm{d}x_1 \ \mathrm{d}x_2 \ \mathrm{d}x_3. \tag{1.7}$$

Applying Newton's second law to the fluid in $V(t)$, the rate of change of momentum

$$\frac{\mathrm{d}}{\mathrm{d}t}\left\{\int_{V(t)} \rho v_i \ \mathrm{d}x_1 \ \mathrm{d}x_2 \ \mathrm{d}x_3\right\},$$

equals the surface forces transmitted across the boundary δV, from the region outside V and the body forces, for example, gravity. We can therefore write

$$\frac{\mathrm{d}}{\mathrm{d}t}\left\{\int_{V(t)} \rho v_i \ \mathrm{d}x_1 \ \mathrm{d}x_2 \ \mathrm{d}x_3\right\} = \int_{\delta V(t)} \sigma_{ij} n_j \, \mathrm{d}S(\boldsymbol{x}) + \int_{V(t)} F_i \ \mathrm{d}x_1 \ \mathrm{d}x_2 \ \mathrm{d}x_3. \tag{1.8}$$

Where $\sigma_{ij} n_j$ is the ith component of stress perpendicular to the area element $\mathrm{d}S(\boldsymbol{x})$ and F_i is the ith component of the body force.

Theorem 1.5.3 (Divergence Theorem)**.** *For a volume $V(t)$, with boundary $\delta V(t)$ we have that*

$$\int_{V(t)} \frac{\partial p_j}{\partial x_j} \ \mathrm{d}x_1 \ \mathrm{d}x_2 \ \mathrm{d}x_3 = \int_{\delta V(t)} p_j n_j \, \mathrm{d}S(\boldsymbol{x}).$$

This is often expressed in vector notation as

$$\int_{V(t)} div p \ \mathrm{d}x_1 \ \mathrm{d}x_2 \ \mathrm{d}x_3 = \int_{\delta V(t)} \mathbf{p} \cdot \mathbf{n} \, \mathrm{d}S(\boldsymbol{x}).$$

Applying the divergence theorem to the second term in (1.8) gives

$$\int_{\delta V(t)} \sigma_{ij} n_j \, \mathrm{d}S(\boldsymbol{x}) = \int_{V(t)} \frac{\partial \sigma_{ij}}{\partial x_j} \ \mathrm{d}x_1 \ \mathrm{d}x_2 \ \mathrm{d}x_3. \tag{1.9}$$

Substituting (1.9) and (1.7) into (1.8) yields

$$\int_{V(t)} \rho \frac{\mathrm{D}v_i}{\mathrm{D}t} \ \mathrm{d}x_1 \ \mathrm{d}x_2 \ \mathrm{d}x_3 = \int_{V(t)} \left(\frac{\partial \sigma_{ij}}{\partial x_j} + F_i \right) \ \mathrm{d}x_1 \ \mathrm{d}x_2 \ \mathrm{d}x_3$$

Since $V(t)$ is an arbitrary volume, we conclude that the integrands must be equal and thus the *Eulerian Equation of Motion* is given by

$$\boxed{\rho \frac{\mathrm{D}v_i}{\mathrm{D}t} = \frac{\partial \sigma_{ij}}{\partial x_j} + F_i.}$$

In elasticity theory, the stress tensor σ_{ij} is usually given as a function of the Lagrangian Coordinates, so

$$\begin{aligned} \frac{\partial \sigma_{ij}}{\partial X_j} &= \frac{\partial \sigma_{ij}}{\partial x_k} \frac{\partial x_k}{\partial X_j}, \\ &= \frac{\partial \sigma_{ij}}{\partial x_k} \frac{\partial}{\partial X_j}(X_k + U_k), && \text{Using } x_k = X_k + U_k \\ &= \frac{\partial \sigma_{ij}}{\partial x_k} \left[\delta_{kj} + \frac{\partial U_k}{\partial X_j} \right]. \end{aligned}$$

If $\left| \dfrac{\partial U_k}{\partial X_j} \right| \ll 1$, then $\dfrac{\partial \sigma_{ij}}{\partial X_j} \approx \dfrac{\partial \sigma_{ij}}{\partial x_j}$. Similarly, density ρ is constant and acceleration $\dfrac{\mathrm{d}v_i}{\mathrm{d}t} = \dfrac{\partial^2 U_i}{\partial t^2}$.

For the small displacement gradient assumption, the *Lagrangian equation of motion* is

$$\boxed{\rho \frac{\partial^2 U_i}{\partial t^2} = \frac{\partial \sigma_{ij}}{\partial X_j} + F_i.}$$

Chapter 2

Linear Elasticity

A constitutive equation in linear (isothermal) isotropic elasticity is required to link stress and strain. These are typically deduced from real world experimental result. In Linear Elasticity we use Hookes' Law.

Definition 2.0.1 (Hooke's Law)**.**

$$\boxed{\sigma_{ij} = \lambda\delta_{ij}E_{KK} + 2\mu E_{ij}}$$

where λ *and* μ *are known as* Lamé Constants.

The Fifteen equations in fifteen unknowns σ_{ij}, E_{ij} and U_i are

$$\rho\frac{\partial^2 U_i}{\partial t^2} = \frac{\partial \sigma_{ij}}{\partial X_j} + F_i \qquad \text{equilibrium equations}$$

$$\sigma_{ij} = \lambda\delta_{ij}E_{kk} + 2\mu E_{ij} \qquad \text{Hooke's Law}$$

$$E_{ij} = \frac{1}{2}\left(\frac{\partial U_i}{\partial X_j} + \frac{\partial U_j}{\partial X_i}\right) \qquad \text{Lagrangian Strain Tensor}$$

with suitable boundary, compatibility and initial conditions.

2.1 An Alternative Expression for E_{ij}

We have the stress tensor in terms of the Strain Tensor, by Hooke's Law

$$\sigma_{ij} = \lambda\delta_{ij}E_{KK} + 2\mu E_{ij}. \tag{2.1}$$

Letting $i = j$, then

$$\begin{aligned} \sigma_{ii} &= \lambda\delta_{ii}E_{KK} + 2\mu E_{ii}, \\ &= (3\lambda + 2\mu)E_{KK}, \\ \Rightarrow \quad E_{KK} &= \frac{\sigma_{ii}}{3\lambda + 2\mu}. \end{aligned}$$

so substituting this into (2.1) yields

$$\sigma_{ij} = \frac{\lambda \delta_{ij} \sigma_{KK}}{3\lambda + 2\mu} + 2\mu E_{ij},$$
$$\therefore \quad E_{ij} = \frac{\sigma_{ij}}{2\mu} - \frac{\lambda}{2\mu(3\lambda + 2\mu)} \delta_{ij} \sigma_{KK}.$$

2.2 An Overview

2.2.1 Formulation of an elastic problem

In general, a problem of elasticity consists of finding the six components of the strain tensor, the six components of the stress tensor and the three displacements in an elastic body subject to surface forces, surface displacement and body forces. We need fifteen equations for these fifteen unknowns. There are three equations of equilibrium, six stress-strain relations and six stain -displacement relation. To ensure a unique value of the displacement components at each point, the strain must also satisfy the compatibility conditions.

2.2.2 Boundary conditions

If the unit normal to the boundary of an elastic body is given by $\boldsymbol{n} = (n_1, n_2, n_3)^T$, then the stress components normal to the surface are given by $\sigma_{nj} = \sigma_{ij} n_i$. It is common to encounter a stress free boundary in which case $\sigma_{nj} = 0$ for all k. Other common boundary conditions include prescribing a displacements os symmetry conditions.

2.2.3 Principle of Superposition

The equations, boundary conditions and compatibility conditions are linear, so for an elastic body the principle of superposition holds. Let the stresses in an elastics body subject to surface forces Q_i and the body forces F_i be σ_{mn}. Let the stresses in the same elastic body subjected to surface force Q_i' and the body forces F_i' be σ_{mn}'. Then $\sigma_{mn} + \sigma_{mn}'$ will represent the stresses due to the surface forces $Q_i + Q_i'$ and body forces $F_i + F_i'$.

2.3 St. Venant Principle

If one applies to a small part of the surface a set of forces which are statically equivalent to zero, then this system will not noticeably effect parts of the body lying away from the origin.

Alternatively if a set of forces acting on a small part of the body is replaced by a system of forces (acting on the same part) which is statically equivalent to the former, then such replacement does not effect parts of the body which do not lie too near the small part of the body

2.4 One-dimensional elasticity

We shall assume that body forces are negligible. There are two cases to consider; one-dimensional state of stress (uniaxial tension) and one-dimensional state of deformation.

2.4.1 Uniaxial Tension

This corresponds to the stress tensor with components:

$$\begin{pmatrix} \sigma_{11} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

where σ_{11} is a non-zero function of X_1. There is no time dependence and the assumption is that the problem is steady.

The three equations of equilibrium are satisfied if $\sigma_{11} = T = \text{constant}$.
We have seen earlier that $\sigma_{ij} = \lambda\delta_{ij}E_{kk} + 2\mu E_{ij}$, gives us that $E_{ij} = \dfrac{1}{2\mu}\sigma_{ij} - \dfrac{\lambda}{2\mu(3\lambda+2\mu)}\delta_{ij}\sigma_{kk}$.
In this case we have

$$\begin{aligned} E_{11} &= \frac{1}{2\mu}\sigma_{11} - \frac{\lambda}{2\mu(3\lambda+2\mu)}\sigma_{11}, \\ &= \frac{(\lambda+\mu)\sigma_{11}}{\mu(3\lambda+2\mu)} = \frac{(\lambda+\mu)T}{\mu(3\lambda+2\mu)}, \\ E_{22} = E_{33} &= -\frac{\lambda\sigma_{11}}{2\mu(3\lambda+2\mu)} = -\frac{\lambda T}{2\mu(3\lambda+2\mu)}, \\ E_{12} &= E_{13} = E_{23} = 0. \end{aligned}$$

Definition 2.4.1.

$$E = \frac{\sigma_{11}}{E_{11}} = \frac{\mu(3\lambda+2\mu)}{\lambda+\mu} \qquad E \text{ is Young's Modulus}$$

$$\nu = -\frac{E_{22}}{E_{11}} = \frac{\lambda}{2(\lambda+\mu)} \qquad \nu \text{ is Poisson's Ratio}$$

Solving for the Lamé constants, give

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)} \quad \mu = \frac{E}{2(1+\nu)}$$

The uniaxial tension test will determine E, ν, λ and ν by conducting experiments.

2.4.2 One-dimensional state of deformation

Consider the case when at all point in the body $U_1 = U_1(X_1)$, $U_2 = U_3 = 0$, we see that $E_{11} = \dfrac{\mathrm{d}U_1}{\mathrm{d}X_1}$, otherwise $E_{ij} = 0$. $\sigma_{11} = \lambda E_{11} + 2\mu E_{11} = (2\mu + \lambda)E_{11}$ and $\sigma_{22} = \sigma_{33} = \lambda E_{11}$.
The equations of equilibrium gives (neglecting body forces).

$$\frac{\mathrm{d}\sigma_{11}}{\mathrm{d}X_1} = 0$$

which implies that σ_{11} is a constant, so E_{11} is a constant and thus $U_1 = E_{11}X_1 + A$, where A is a constant of integration which represents the rigid body motion.

2.5 Plane Elasticity

If all planes initially normal to the X_3 axis remain normal to it after deformation and all straight lines initially parallel to the X_3 axis remain parallel to it after deformation then a state of plane deformation exists. These conditions are expressed by

$$U_1 = U_1(X_1, X_2); \quad U_2 = (X_1, X_2); \quad U_3 = U_3(X_3).$$

Plane Strain
The state of plane strain corresponds to the conditions

$$U_1 = U_1(X_1, X_2); \quad U_2 = (X_1, X_2); \quad U_3 = 0.$$

Plane Stress
The state of plane stress in a body is such that

$$\sigma_{11} = \sigma_{11}(X_1, X_2); \quad \sigma_{22} = \sigma_{22}(X_1, X_2); \quad \sigma_{12} = \sigma_{12}(X_1, X_2)$$
$$\sigma_{13} = \sigma_{23} = \sigma_{33} = 0.$$

2.5.1 Plane Strain

In plane strain we have

$$E_{11} = \frac{\partial U_1}{\partial X_1}, \quad E_{22} = \frac{\partial U_2}{\partial X_2}, \quad E_{12} = \frac{1}{2}\left(\frac{\partial U_1}{\partial X_2} + \frac{\partial U_2}{\partial X_1}\right).$$
$$E_{13} = E_{23} = E_{33} = 0.$$

The stress-strain relation becomes

$$\begin{aligned}
\sigma_{11} &= 2\mu E_{11} + \lambda(E_{11} + E_{22}),\\
\sigma_{22} &= 2\mu E_{22} + \lambda(E_{11} + E_{22}),\\
\sigma_{12} &= 2\mu E_{12},\\
\sigma_{33} &= \lambda(E_{11} + E_{22}),\\
\sigma_{13} &= \sigma_{13} = 0.
\end{aligned}$$

The equations for equilibrium are

$$\frac{\partial \sigma_{11}}{\partial X_1} + \frac{\partial \sigma_{21}}{\partial X_2} + F_1 = 0; \quad \frac{\partial \sigma_{12}}{\partial X_1} + \frac{\partial \sigma_{22}}{\partial X_2} + F_2 = 0. \tag{2.2}$$

Five (of the six) compatibility conditions are satisfied automatically, and we retain

$$2\frac{\partial^2 E_{12}}{\partial X_1 \partial X_2} = \frac{\partial^2 E_{11}}{\partial X_2^2} + \frac{\partial^2 E_{22}}{\partial X_1^2}. \tag{2.3}$$

The Stress boundary conditions are given(using $T_j = \sigma_{ij} n_i$) by

$$\sigma_{n1} = n_1 \sigma_{11} + n_2 \sigma_{21},$$
$$\sigma_{n2} = n_1 \sigma_{12} + n_2 \sigma_{22}.$$

2.5.2 Compatibility conditions for stress

Consider a linear elastic body in a state of plane strain with constant body force. Differentiating the first equation in (2.2) with respect to X_1 and the second equation with respect to X_2 gives

$$\frac{\partial^2 \sigma_{11}}{\partial X_1^2} + \frac{\partial^2 \sigma_{12}}{\partial X_1 \partial X_2} = 0; \quad \frac{\partial^2 \sigma_{12}}{\partial X_1 \partial X_2} + \frac{\partial^2 \sigma_{22}}{\partial X_2^2} = 0.$$

Adding these two equations gives

$$2\frac{\partial^2 \sigma_{12}}{\partial X_1 \partial X_2} + \frac{\partial^2 \sigma_{11}}{\partial X_1^2} + \frac{\partial^2 \sigma_{22}}{\partial X_2^2} = 0.$$

Substituting for σ_{ij}'s from the stress-strain relations yields

$$4\mu \frac{\partial^2 E_{12}}{\partial X_1 \partial X_2} + 2\mu \left(\frac{\partial^2 E_{11}}{\partial X_1^2} + \frac{\partial^2 E_{22}}{\partial X_2^2} \right) + \lambda \left(\frac{\partial^2}{\partial X_1^2} + \frac{\partial^2}{\partial X_2^2} \right) (E_{11} + E_{22}) = 0.$$

Using the retained compatibility conditions (2.3) yields

$$2\mu \left(\frac{\partial^2}{\partial X_1^2} + \frac{\partial^2}{\partial X_2^2} \right) (E_{11} + E_{22}) + \lambda \left(\frac{\partial^2}{\partial X_1^2} + \frac{\partial^2}{\partial X_2^2} \right) (E_{11} + E_{22}) = 0.$$

Since in general $\lambda + 2\mu \neq 0$ we conclude that $\left(\frac{\partial^2}{\partial X_1^2} + \frac{\partial^2}{\partial X_2^2} \right) (E_{11} + E_{22}) = 0.$

Adding stress-strain relations for σ_{11} and σ_{22} and differentiating we can see that

$$\sigma_{11} + \sigma_{22} = 2(\mu + \lambda)(E_{11} + E_{22})$$

Substituting this in the above result gives that

$$\left(\frac{\partial^2}{\partial X_1^2} + \frac{\partial^2}{\partial X_2^2}\right)(\sigma_{11} + \sigma_{22}) = 0. \tag{2.4}$$

Notice that equations (2.2) and (2.4) are three equations in the three unknowns $\sigma_{11}, \sigma_{22}, \sigma_{12}$.

Example 2.5.1. *A uniform linearly elasticity isotropic material of rectangular cross section and thickness $2h$ is fixed at $X_1 = 0$ to a rigid wall is bent by a shear force P suitably distributed at $X_1 = l$ where l is the length The static conservation of momentum equations without body forces*

are

$$\frac{\partial \sigma_{11}}{\partial X_1} + \frac{\partial \sigma_{12}}{\partial X_2} = 0; \quad \frac{\partial \sigma_{12}}{\partial X_1} + \frac{\partial \sigma_{22}}{\partial X_2} = 0.$$

The retained compatibility condition is

$$\left(\frac{\partial^2}{\partial X_1^2} + \frac{\partial^2}{\partial X_2^2}\right)(\sigma_{11} + \sigma_{22}) = 0.$$

The boundary conditions are that

Top Surface $\boldsymbol{n} = (0, 1, 0)$ *so* $\sigma_{21}(X_1, h) = \sigma_{12}(X_1, h) = 0$ *by symmetry and* $\sigma_{22}(X_1, h) = 0$.

Bottom Surface $\boldsymbol{n} = (0, -1, 0)$ *so* $\sigma_{21}(X_1, -h) = 0$ *and* $\sigma_{22}(X_1, -h) = 0$.

Shear Surface $\boldsymbol{n} = (1, 0, 0)$, *now* $\sigma_{11}(l, X_2) = 0$ *and*

$$-P = \int_{X_2=-h}^{h} \sigma_{12}(l, X_2)\, \mathrm{d}X_2.$$

The solution procedure is as follows

1. *We have 3 equations in 3 unknowns for stress, so solve or stress.*
2. *Use the Stress-strain relations to determine the strains.*
3. *Use the strain-displacement equations to determine the displacements (We do not need to apply the compatibility conditions in this case).*

2.5.3 Plane Stress

The Stress-strain relations become

$$\begin{aligned}
\sigma_{33} = 0 \quad &\Rightarrow \quad E_{33} = \frac{-\lambda(E_{11}+E_{22})}{2\mu+\lambda} \\
\sigma_{11} &= 2\mu E_{11} + \lambda E_{KK} = 2\mu E_{11} + \frac{2\mu\lambda}{2\mu+\lambda}(E_{11}+E_{22}) \\
\sigma_{22} &= 2\mu E_{22} + \lambda E_{KK} = 2\mu E_{22} + \frac{2\mu\lambda}{2\mu+\lambda}(E_{11}+E_{22}) \\
\sigma_{12} &= 2\mu E_{12}
\end{aligned}$$

Equations for equilibrium

$$\frac{\partial \sigma_{11}}{\partial X_1} + \frac{\partial \sigma_{12}}{\partial X_2} + F_1 = 0, \qquad \frac{\partial \sigma_{12}}{\partial X_1} + \frac{\partial \sigma_{22}}{\partial X_2} + F_2 = 0.$$

Of the six compatibility conditions two are automatically satisfied leaving

$$2\frac{\partial^2 E_{12}}{\partial X_1 \partial X_2} = \frac{\partial^2 E_{11}}{\partial X_2^2} + \frac{\partial^2 E_{22}}{\partial X_1^2}, \qquad \frac{\partial^2 E_{33}}{\partial X_1^2} = \frac{\partial^2 E_{33}}{\partial X_2^2} = \frac{\partial^2 E_{33}}{\partial X_1 \partial X_2} = 0.$$

The stress boundary conditions becomes

$$T_1 = n_1\sigma_{11} + n_2\sigma_{12}.$$

$$T_2 = n_1\sigma_{12} + n_2\sigma_{22}.$$

where $\boldsymbol{n} = (n_1, n_2, 0)^T$ is the unit normal.

2.6 A Three Dimensional Problem - Hydrostatic Pressure

$$U_1 = \alpha X_1, \quad U_2 = \alpha X_2, \quad U_2 = \alpha X_3$$

Hence $E_{11} = E_{22} = E_{33} = \alpha$ and $E_{12} = E_{13} = E_{23} = 0$ so we have pure dilatation. Thus

$$\begin{aligned}
\sigma_{11} = \sigma_{22} = \sigma_{33} &= (3\lambda + 2\mu)\alpha \\
\sigma_{12} = \sigma_{23} = \sigma_{13} &= 0
\end{aligned}$$

The bulk modulus is $\lambda + \frac{2}{3}\mu$. We need $\lambda + \frac{2\mu}{3} > 0$, so

$$\begin{aligned}
\lambda + \frac{2\mu}{3} &= \frac{E\nu}{(1+\nu)(1-2\nu)} + \frac{2}{3}\frac{E}{2(1+\nu)}, \\
&= \frac{E}{1+\nu}\left[\frac{\nu}{1-2\nu} + \frac{1}{3}\right], \\
&= \frac{E(3\nu+1-2\nu)}{3(1+\nu)(1-2\nu)}, \\
&= \frac{E}{3(1-2\nu)} > 0.
\end{aligned}$$

so we conclude that $\nu \leqslant \frac{1}{2}$.

If we rewrite the stress-strain relation in terms of Young's Modulus and Poisson's ratio

$$E_{ij} = \frac{1+\nu}{E}\sigma_{ij} - \frac{\nu}{E}\delta_{ij}\sigma_{kk},$$

and let $\sigma_{ij} = -p\delta_{ij}$ the hydrostatic pressure, then

$$E_{11} = E_{22} = E_{33} = -\frac{(1+\nu)}{E}p + \frac{3\nu}{E}p = -\frac{p}{E}(1-2\nu).$$

If $\nu = 1/2$, then there is no volume change, thus the material is incompressible. For metals $\nu \sim 0.3$.

2.7 Torsion

Consider a bar twisted by a couple applied at the base. We have to determine a solution of the equations of linear elasticity with zero stress on lateral surfaces and moments equivalent to a couple applied on the base. The St Venant principle says that this will be the solution except near the end. The equations to be satisfied are

$$\frac{\partial \sigma_{ij}}{\partial x_j} = 0, \qquad \sigma_{ij} = \lambda\delta_{ij}E_{kk} + 2\mu E_{ij}.$$

$$E_{ij} = \frac{1}{2}\left(\frac{\partial U_i}{\partial X_j} + \frac{\partial U_j}{\partial X_i}\right).$$

with suitable boundary conditions.

$$X_1 = r\cos\alpha \qquad X_2 = r\sin\alpha$$
$$x_1 = r\cos(\theta+\alpha) \qquad x_2 = r\sin(\theta+\alpha)$$

where $\theta \ll 1$. From the definition of U_1 and at leading order we have[1].

$$\begin{aligned} U_1 &= r\cos(\theta+\alpha) - r\cos\alpha \\ &= \theta r\left[\frac{\cos(\theta+\alpha)-\cos(\alpha)}{(\theta+\alpha)-\alpha}\right] \\ &\approx -\theta r\sin\alpha \\ &= -\theta X_2 \\ U_2 &\approx \theta X_1 \end{aligned}$$

Now $\theta = \tau X_3$, where τ is the twist per unit length.

$$U_1 = -\tau X_2 X_3; \qquad U_2 = \tau X_1 X_3.$$

We shall now consider two cases, firstly a cylindrical bar and secondly a bar of arbitrary cross section.

2.7.1 A circular cylinder, with radius a.

We guess the solution as $U_1 = -\tau X_2 X_3$, $U_2 = \tau X_1 X_3$, $U_3 = 0$, now this yields

$$\begin{aligned} \sigma_{11} = \sigma_{22} = \sigma_{33} = \sigma_{12} = 0, \\ \sigma_{12} = -\mu\tau X_2, \qquad \sigma_{23} = \mu\tau X_1. \end{aligned}$$

We note $\dfrac{\partial \sigma_{ij}}{\partial X_j} = 0$ for all j.

The Lateral surface is $X_1^2 + X_2^2 - a^2 = 0$, has unit normal $\boldsymbol{n} = (\frac{X_1}{a}, \frac{X_2}{a}, 0)$. Now we use $\sigma_{nj} = \sigma ij n_j$, so give

$$\begin{aligned} \sigma_{n1} &= \sigma_{11}n_1 + \sigma_{21}n_2 + \sigma_{31}n_3 = 0 \\ \sigma_{n2} &= \sigma_{12}n_1 + \sigma_{22}n_2 + \sigma_{32}n_3 = 0 \\ \sigma_{n3} &= \sigma_{13}n_1 + \sigma_{23}n_2 + \sigma_{33}n_3 = -\mu\tau X_2\frac{X_1}{a} + \mu\tau X_1\frac{X_2}{a} = 0 \end{aligned}$$

Thus our guess is validated as a solution.

[1] An alternative to the stated method is to use the double-angle formulae

2.7.2 Arbitrary cross section

The displacement $U_1 = -\tau X_2 X_3$, $U_2 = \tau X_1 X_3$ and $U_3 = 0$ still satisfy the equations of equilibrium but not the boundary conditions. Now consider displacement out of the X_1, X_2 plane such as

$$U_1 = -\tau X_2 X_3, U_2 = \tau X_1 X_3, U_3 = \tau\phi(X_1, X_2),$$

where $\phi(X_1, X_2)$ must be determined.

Now $E_{11} = E_{22} = E_{23} = E_{12} = 0$, but

$$E_{13} = \frac{1}{2}\left(-\tau X_2 + \tau\frac{\partial\phi}{\partial X_1}\right),$$
$$E_{23} = \frac{1}{2}\left(\tau X_1 + \tau\frac{\partial\phi}{\partial X_2}\right).$$

Also $\sigma_{11} = \sigma_{22} = \sigma_{33} = \sigma_{12} = 0$, and

$$\sigma_{13} = \tau\mu\left(\frac{\partial\phi}{\partial X_1} - X_2\right),$$
$$\sigma_{23} = \tau\mu\left(\frac{\partial\phi}{\partial X_2} + X_1\right).$$

Now, applying the Lagrangian description of motion we see

$$\frac{\partial\sigma_{11}}{\partial X_1} + \frac{\partial\sigma_{12}}{\partial X_2} + \frac{\partial\sigma_{13}}{\partial X_3} = 0,$$
$$\frac{\partial\sigma_{21}}{\partial X_1} + \frac{\partial\sigma_{22}}{\partial X_2} + \frac{\partial\sigma_{23}}{\partial X_3} = 0,$$
$$\frac{\partial\sigma_{31}}{\partial X_1} + \frac{\partial\sigma_{32}}{\partial X_2} + \frac{\partial\sigma_{33}}{\partial X_3} = \tau\mu\left(\frac{\partial^2\phi}{\partial X_1^2} + \frac{\partial^2\phi}{\partial X_2^2}\right) = 0.$$

thus we require that ϕ is a solution of Laplace's equation

$$\frac{\partial^2\phi}{\partial X_1^2} + \frac{\partial^2\phi}{\partial X_2^2} = 0,$$

We apply the no stress boundary conditions on the lateral surfaces. Let $\boldsymbol{n} = (n_1, n_2, 0)^T$, be the unit normal then

$$\sigma_{n1} = \sigma_{11}n_1 + \sigma_{12}n_2 = 0$$
$$\sigma_{n2} = \sigma_{12}n_1 + \sigma_{22}n_2 = 0$$
$$\sigma_{n3} = \sigma_{13}n_1 + \sigma_{23}n_3 = \tau\mu\left[n_1\left(\frac{\partial\phi}{\partial X_1}\right) + n_2\left(\frac{\partial\phi}{\partial X_2} + X_1\right)\right] = \tau\mu[\boldsymbol{n}.\nabla\phi + \boldsymbol{n}.(-X_2, X_1, 0)] = 0$$

The problem has thus been reduced to solving the partial-differential equation

$$\frac{\partial^2 \phi}{\partial X_1^2} + \frac{\partial^2 \phi}{\partial X_2^2} = 0,$$

subject to the boundary conditions that

$$\boldsymbol{n}.\nabla\phi = \boldsymbol{n}.(X_2, -X_1, 0).$$

We consider

$$\begin{aligned}\iint_A \sigma_{12}\,\mathrm{d}X_1\,\mathrm{d}X_2 &= \mu\tau \iint_A \left(\frac{\partial \phi}{\partial X_1} - X_2\right)\,\mathrm{d}X_1\,\mathrm{d}X_2 \\ &= \mu\tau \iint_A \frac{\partial}{\partial X_1}\left[X_1\left(\frac{\partial \phi}{\partial X_1} - X_2\right)\right] + \frac{\partial}{\partial X_2}\left[X_1\left(\frac{\partial \phi}{\partial X_2} - X_1\right)\right]\,\mathrm{d}X_1\,\mathrm{d}X_2\end{aligned}$$

Applying the Divergence Theorem

$$= \mu\tau \int_C \left(X_1\left(\frac{\partial \phi}{\partial X_1} - X_2\right), X_1\left(\frac{\partial \phi}{\partial X_2} + X_1\right)\right).(n_1, n_2)\,\mathrm{d}l$$

Applying the BC's

$$= \mu\tau X_1 \int_C [\boldsymbol{n}.\nabla\phi + \boldsymbol{n}.(-X_2, X_1, 0)]\,\mathrm{d}l = 0$$

Similarly $\iint_A \sigma_{23}\,\mathrm{d}X_1\,\mathrm{d}X_2 = 0$, so there are no resultant forces. The resultant is a couple.

Chapter 3

Waves and Fluids

3.1 The Navier Equations

We have the time dependent equations of linear elasticity,

$$\rho \frac{\partial^2 U_i}{\partial t^2} = \frac{\partial \sigma_{ij}}{\partial X_j} + F_i, \tag{3.1}$$

$$\sigma_{ij} = \lambda \delta_{ij} E_{KK} + 2\mu E_{ij}, \tag{3.2}$$

$$E_{ij} = \frac{1}{2}\left(\frac{\partial U_i}{\partial X_j} + \frac{\partial U_j}{\partial X_i}\right). \tag{3.3}$$

Substituting (3.2) into (3.1), gives

$$\begin{aligned}
\rho \frac{\partial^2 U_i}{\partial t^2} &= \frac{\partial}{\partial X_j}(\lambda \delta_{ij} E_{KK} + 2\mu E_{ij}) + F_i, \\
&= \lambda \frac{\partial}{\partial X_i} E_{KK} + 2\mu \frac{\partial}{\partial X_j} E_{ij} + F_i, \\
&= \lambda \frac{\partial}{\partial X_i}\left(\frac{\partial U_K}{\partial X_K}\right) + 2\mu \frac{\partial}{\partial X_j}\left\{\frac{1}{2}\left(\frac{\partial U_i}{\partial X_j} + \frac{\partial U_j}{\partial X_i}\right)\right\} + F_i, \qquad \text{Using (3.3)} \\
&= \lambda \frac{\partial}{\partial X_i}\left(\frac{\partial U_K}{\partial X_K}\right) + \mu \frac{\partial^2 U_i}{\partial X_j^2} + \mu \frac{\partial}{\partial X_i}\left(\frac{\partial U_j}{\partial X_j}\right) + F_i, \\
&= (\lambda + \mu) \frac{\partial}{\partial X_i}\left(\frac{\partial U_K}{\partial X_K}\right) + \mu \frac{\partial^2 U_i}{\partial X_j^2} + F_i.
\end{aligned}$$

In vector notation this may be written more succinctly as

$$\rho \frac{\partial^2 \boldsymbol{U}}{\partial t^2} = (\lambda + \mu)\nabla(\nabla.\boldsymbol{U}) + \mu \nabla^2 \boldsymbol{U} + \boldsymbol{F}.$$

Using the identity $\nabla \times (\nabla \times \boldsymbol{U}) = \nabla(\nabla.\boldsymbol{U}) - \nabla^2 \boldsymbol{u}$, we obtain, an alternative form for the Navier Equations, namely

$$\rho \frac{\partial \boldsymbol{U}}{\partial t} = (\lambda + 2\mu)\nabla(\nabla.\boldsymbol{U}) - \mu \nabla \times (\nabla \times \boldsymbol{U}) + \boldsymbol{F}.$$

The standard form of the Navier Equations are:

$$\rho\frac{\partial^2 U_i}{\partial t^2} = (\lambda+\mu)\frac{\partial}{\partial X_i}\left(\frac{\partial U_K}{\partial X_K}\right) + \mu\frac{\partial^2 U_i}{\partial X_j^2} + F_i, \tag{3.4}$$

$$\rho\frac{\partial^2 \boldsymbol{U}}{\partial t^2} = (\lambda+\mu)\nabla(\nabla.\boldsymbol{U}) + \mu\nabla^2\boldsymbol{U} + \boldsymbol{F}.$$

These are wave equations, for example modelling the vibration in the panels of a car or devastation caused by seismic activity.

3.1.1 Waves in Elastic Solids

We assume body forces are negligible and so we have $\boldsymbol{F} = 0$.

- Differentiating the Navier Equations with respect to X_i gives

$$\rho\frac{\partial^2}{\partial t^2}\left(\frac{\partial U_i}{\partial X_i}\right) = (\lambda+\mu)\frac{\partial^2}{\partial X_i^2}\left(\frac{\partial U_K}{\partial X_K}\right) + \mu\frac{\partial^2}{\partial X_j^2}\left(\frac{\partial U_i}{\partial X_i}\right),$$
$$= (\lambda+2\mu)\frac{\partial^2}{\partial X_i^2}\left(\frac{\partial U_K}{\partial X_K}\right).$$

  Notice that the *dilatation* $= \Delta = E_{KK} = \dfrac{\partial U_K}{\partial X_K}$, thus

$$\frac{\partial^2\Delta}{\partial t^2} = \frac{\lambda+2\mu}{\rho}\frac{\partial^2\Delta}{\partial X_i^2} = C_1^2\frac{\partial^2\Delta}{\partial X_i^2}$$

  These are waves associated with dilatation (volume change) and travel with a speed $C_1 = \sqrt{(\lambda+2\mu)/\rho}$.

- The Navier Equation

$$\rho\frac{\partial^2 U_i}{\partial t^2} = (\lambda+\mu)\frac{\partial}{\partial X_i}\left(\frac{\partial U_k}{\partial X_k}\right) + \mu\frac{\partial^2 U_i}{\partial X_k^2}, \tag{3.5}$$

  can also be written as (interchanging i and j)

$$\rho\frac{\partial^2 U_j}{\partial t^2} = (\lambda+\mu)\frac{\partial}{\partial X_j}\left(\frac{\partial U_m}{\partial X_m}\right) + \mu\frac{\partial^2 U_j}{\partial X_k^2}. \tag{3.6}$$

  We shall take $\dfrac{\partial}{\partial X_j}(3.5) - \dfrac{\partial}{\partial X_i}(3.6)$ which gives

$$\rho\frac{\partial^2}{\partial t^2}\left(\frac{\partial U_i}{\partial X_j} - \frac{\partial U_j}{\partial X_i}\right) = (\lambda+\mu)\frac{\partial^2}{\partial X_i\partial X_j}\left(\frac{\partial U_k}{\partial X_k} - \frac{\partial U_m}{\partial X_m}\right) + \mu\frac{\partial^2}{\partial X_k^2}\left(\frac{\partial U_i}{\partial X_j} - \frac{\partial U_j}{\partial X_i}\right).$$

$$\Rightarrow \qquad \frac{\partial^2 R_{ij}}{\partial t^2} = \frac{\mu}{\rho}\frac{\partial^2 R_{ij}}{\partial X_k^2} = C_2^2\frac{\partial^2 R_{ij}}{\partial X_k^2}.$$

where

$$R_{ij} = \frac{1}{2}\left(\frac{\partial U_i}{\partial X_j} - \frac{\partial U_j}{\partial X_i}\right).$$

These are waves associated with rotation and travel at a speed $C_2 = \sqrt{\mu/\rho}$.

Example 3.1.1 (One-dimensional dilatation waves)**.** *Consider displacements of the form* $U_1 = U_1(X_1, t)$, $U_2 = U_3 = 0$ *then we have* $\Delta = \dfrac{\partial U_1}{\partial X_1}$ *and* $\boldsymbol{\omega} = 0$. *The dilatation obeys the one-dimensional wave equation, that is*

$$\frac{\partial^2 \Delta}{\partial t^2} = C_1^2 \frac{\partial^2 \Delta}{\partial X_1^2}.$$

The solution to this is known as D'Alembert's solution namely

$$\Delta = f(X_1 - C_1 t) + g(X_1 + C_1 t)$$

Example 3.1.2 (One-dimensional rotation waves)**.**

Consider $U_2 = U_2(X_1, t)$, $U_1 = U_3 = 0$ *thus* $\Delta = 0$ *and*

$$R_{12} = -\frac{1}{2}\frac{\partial U_2}{\partial X_1}.$$

The single non-zero component of the rotation tensor satisfies the one-dimensional wave equation, with speed C_2. *These waves are often called shear waves.*

3.2 Fluids

For fluids the independent variables are the Eulerian coordinates. We have the equations for conservation of mass

$$\frac{\partial \rho}{\partial t} + v_i \frac{\partial \rho}{\partial x_i} + \rho \frac{\partial v_k}{\partial x_k} = 0,$$

and for the conservation of momentum

$$\rho\left(\frac{\partial v_i}{\partial t} + v_l \frac{\partial v_i}{\partial x_l}\right) = \frac{\partial \sigma_{ij}}{\partial x_j} + F_i.$$

We however require an additional equation.

3.2.1 Stress Tensor in Fluids

For elasticity we related stress to strain via Hooke's Law, this is a very poor model for fluids. Instead we relate stress to the deformation rate tensor

$$2d_{kl} = \frac{\partial v_k}{\partial x_l} + \frac{\partial v_l}{\partial x_k}$$

A NEWTONIAN liquid is one which is linear in the velocity gradient. We assume for such fluids that

$$\sigma_{ij} = \alpha\delta_{ij} - \frac{2}{3}\beta d_{kk}\delta_{ij} + 2\beta d_{ij}$$

Consider a frictionless fluid, with no shear forces, then the only components of σ_{ij} which are non-zero are σ_{11}, σ_{22} and σ_{33}. The pressure in a liquid is the same in all direction, so $\sigma_{11} = \sigma_{22} = \sigma_{33} = -p$, where p is the pressure. Thus for a frictionless fluid, which we term inviscid

$$\boxed{\sigma_{ij} = -p\delta_{ij}.}$$

For a viscous fluid (that is one with fiction), we defined pressure

$$p = -\frac{1}{3}(\sigma_{11} + \sigma_{22} + \sigma_{33}) = \frac{1}{3}\sigma_{ii}.$$

We recall from Examples Sheet 2 that σ_{kk} is an invariant of the symmetric tensor σ_{ij},

If we substitute the definition of σ_{ij} into this equations we have

$$p = -\frac{1}{3}(\alpha\delta_{ii} - \frac{2}{3}\beta d_{kk}\delta_{ii} + 2\beta d_{ii}) = -\frac{1}{3}(3\alpha - 2\beta d_{kk} + 2\beta d_{kk}),$$

therefore

$$p = -\alpha.$$

Thus

$$\begin{aligned}\sigma_{ij} &= (-p - \frac{2\beta d_{kk}}{3})\delta_{ij} + 2\beta d_{ij},\\ &= -p\delta_{ij} - \frac{2\beta}{3}d_{kk}\delta_{ij} + 2\beta d_{ij}.\end{aligned}$$

β depends on viscous forces and is usually written as μ, thus

$$\boxed{\sigma_{ij} = -p\delta_{ij} - \frac{2\mu}{3}d_{kk}\delta_{ij} + 2\mu d_{ij}.}$$

μ is called the VISCOSITY or sometimes the DYNAMIC SHEAR VISCOSITY.

The constant of proportional is a direct measure of Viscosity.

Material	Viscosity /Nsm^{-2}
Air 0^oC	171×10^{-7}
Water 0^oC	1.79×10^{-3}
Pitch 0^oC	51×10^{9}

3.2.2 Inviscid Flow ($\mu = 0$)

$$\rho\left(\frac{\partial v_i}{\partial t} + v_l\frac{\partial v_i}{\partial x_l}\right) = \frac{\partial \sigma_{ij}}{\partial x_j} + F_i \qquad \sigma_{ij} = -p\delta_{ij}.$$

The Eulerian equation of motion is therefore reduced to

$$\rho\left(\frac{\partial v_i}{\partial t} + v_l\frac{\partial v_i}{\partial x_l}\right) = -\frac{\partial p}{\partial x_i} + F_i.$$

With the conservation of mass equation given by

$$\frac{\partial \rho}{\partial t} + v_l\frac{\partial \rho}{\partial x_l} + \rho\frac{\partial v_k}{\partial x_k} = 0.$$

This is a pde system with four equations in 5 unknowns, and thus to solve we need additional information (eg Conservation of energy and state equations). However if the density ρ is a known constant then we have 4 equation in four unknown. The equations for an inviscid incompressible (constant pressure) fluid are therefore:

$$\frac{\partial v_k}{\partial x_k} = 0, \qquad \rho\left(\frac{\partial v_i}{\partial t} + v_l\frac{\partial v_i}{\partial x_l}\right) = -\frac{\partial p}{\partial x_i} + F_i$$

The boundary condition for a fixed boundary is $\boldsymbol{u}.\boldsymbol{n} = 0$, on that boundary.

3.2.3 Viscous Fluid – The Navier-Stokes Equations

We have the conservation of momentum equation

$$\rho\left(\frac{\partial v_i}{\partial t} + v_l\frac{\partial v_i}{\partial x_l}\right) = \frac{\partial \sigma_{ij}}{\partial x_j} + F_i, \tag{3.7}$$

and the stress deformation rate relation

$$\sigma_{ij} = -p\delta_{ij} - \frac{2\mu}{3}d_{kk}\delta_{ij} + 2\mu d_{ij}. \tag{3.8}$$

The definition of the rate of strain tensor is

$$d_{ij} = \frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right). \tag{3.9}$$

Substituting (3.9) into (3.8) yields

$$\sigma_{ij} = -p\delta_{ij} - \frac{2\mu}{3}\frac{\partial v_k}{\partial x_k}\delta_{ij} + \mu\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right),$$

Differentiating this with respect to x_j gives

$$\begin{aligned}\frac{\partial \sigma_{ij}}{\partial x_j} &= -\frac{\partial p}{\partial x_i} - \frac{2\mu}{3}\frac{\partial}{\partial x_i}\left(\frac{\partial v_k}{\partial x_k}\right) + \mu\frac{\partial^2 v_i}{\partial x_j^2} + \mu\frac{\partial}{\partial x_i}\left(\frac{\partial v_j}{\partial x_j}\right), \\ &= -\frac{\partial p}{\partial x_i} + \frac{\mu}{3}\frac{\partial}{\partial x_i}\left(\frac{\partial v_k}{\partial x_k}\right) + \mu\frac{\partial^2 x_i}{\partial x_j^2}. \end{aligned} \tag{3.10}$$

Substituting (3.10) into (3.7) gives, the *Navier–Stokes Equations.*

$$\boxed{\rho\left(\frac{\partial v_i}{\partial t} + v_l\frac{\partial v_i}{\partial x_l}\right) = -\frac{\partial p}{\partial x_i} + \mu\frac{\partial^2 v_i}{\partial x_j^2} + \frac{\mu}{3}\frac{\partial}{\partial x_i}\left(\frac{\partial v_k}{\partial x_k}\right) + F_i.}$$

To solve a system for a compressible fluid we have four equations in five unknowns (ρ, v_i, p) with the equations for Conservation of mass and the Navier–Stokes Equations namely

$$\frac{\partial \rho}{\partial t} + v_l\frac{\partial \rho}{\partial x_l} + \rho\frac{\partial v_k}{\partial x_l} = 0,$$

$$\rho\left(\frac{\partial v_i}{\partial t} + v_l\frac{\partial v_i}{\partial x_l}\right) = -\frac{\partial p}{\partial x_i} + \mu\frac{\partial^2 v_i}{\partial x_j^2} + \frac{\mu}{3}\frac{\partial}{\partial x_i}\left(\frac{\partial v_k}{\partial x_k}\right) + F_i.$$

To solve such a system requires additional information, derived from the theories of thermodynamics but this is beyond the scope of this course.

If however we have ρ as a known constant, the above equations reduce to four equations in four unknowns

$$\boxed{\frac{\partial v_k}{\partial x_k} = 0, \qquad \rho\left(\frac{\partial v_i}{\partial t} + v_l\frac{\partial v_i}{\partial x_l}\right) = -\frac{\partial p}{\partial x_i} + \mu\frac{\partial^2 v_i}{\partial x_j^2} + F_i.}$$

These are the equations for a viscous, incompressible flow. These equations will be studies in depth in the second half of this course "Continuum Mechanics II – Fluid Mechanics".

The Boundary conditions at a fixed boundary is $\boldsymbol{u} = \mathbf{0}$, this is refereed to as the non-slip boundary conditions and explains why a layer of dust remains on a vehicle even after driving on the motorway. The Navier–Stokes equations are of a higher order than Euler's equations and therefore we require more boundary conditions than in the case of inviscid flow.